# Electronic Supplementary Information

## Conformal campaigns: from coverage requests to audited consequences

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## S1 Notation and the outcome table

The *audit set* of the main text is a labelled random sample of the screened library used neither for training nor for calibration.

Labels are $y \in \{0,1\}$: 1 for the in-class compounds, those with the property the screen looks for (toxic, soluble, or reaching the conductivity target), and 0 for the out-class compounds. The trained model gives predicted probabilities $\hat{p}(y \mid x)$ and nonconformity scores $s(x,y) = 1 - \hat{p}(y \mid x)$. For class $y$ let $S^{(y)}_{(1)} \le \cdots \le S^{(y)}_{(n_y)}$ be the sorted scores of the $n_y$ calibration compounds of class $y$, and $\tau_y = S^{(y)}_{(k_y)}$. The prediction set is $C(x) = \{y : s(x,y) \le \tau_y\}$, one of $\{0\}$, $\{1\}$, $\{0,1\}$, $\varnothing$.

For a fixed pair of thresholds, write $c \in \{\{0\},\{1\},\{0,1\},\varnothing\}$ for the prediction set and

$$p_{y,c} = \Pr\big(Y = y,\ C(X) = c\big) \tag{S1}$$

for the probability of a cell of the outcome table (main text Table 1) when compounds are independent. Under exchangeability we read $p_{y,c}$ as $p_{y,c}(Q)$, the long-run frequency of the cell given the de Finetti distribution $Q$ and the

thresholds (Section S3). Any screening quantity is

$$p_g = \sum_{y,c} g(y,c)\, p_{y,c}, \qquad g(y,c) \in \{0,1\}, \tag{S2}$$

for a fixed indicator $g$ that selects cells. Examples: missed in-class compounds, $g = \mathbf{1}\{y = 1, c = \{0\}\}$; false leads, $g = \mathbf{1}\{y = 0, c = \{1\}\}$; deferred, $g = \mathbf{1}\{c \in \{\{0,1\}, \varnothing\}\}$; coverage, $g = \mathbf{1}\{y \in c\}$. Ratios of such sums, for example the fraction of in-class compounds among compounds labelled $\{1\}$, follow directly. The same identity holds for counts: if $K_{y,c}$ is the number of test compounds in cell $(y,c)$, the count of quantity $g$ is $K_g = \sum_{y,c} g(y,c) K_{y,c}$. Within each class the cells sum to the class total, $\sum_c p_{y,c} = \Pr(Y = y)$.

# S2 Coverage given the calibration set and the finite-sample correction

**Exact laws.** Consider one class and drop the index $y$; let there be $n$ calibration scores and $\tau = S_{(k)}$, $u = n + 1 - k$; in this section $u$ denotes this grid index, not the predicted probability. Scores are assumed continuous. With tied scores, randomized tie-breaking restores all laws in this section exactly: every compound, calibration or future, receives an independent uniform number, and compounds are ordered by score and then by that number. This needs no knowledge of future scores. Our scripts implement it equivalently, on stored scores, by adding a uniform jitter smaller than the smallest gap between distinct scores. Without it, ties keep the one-sided coverage statement conservative, but they can break the two-sided count intervals of Sections S3 and S4: a cluster of tied scores at a threshold moves together, so the upper limit for the number of future compounds at or below the threshold can be exceeded (Section S8, Table S4). We therefore break ties at random in all molecular analyses. For a future batch of $m$ compounds of this class, the number whose score is at most $\tau$ depends only on the ranks of the $n+m$ exchangeable scores, all orderings of which are equally likely. Its distribution is therefore

$$\text{BetaBinomial}(m; k, u), \tag{S3}$$

exactly and for any exchangeable data.[1] If compounds are independent with score distribution $F$, the coverage of the deployed model is $F(S_{(k)})$. This is the $k$-th order statistic of $n$ independent uniform variables, so

$$F(S_{(k)}) \sim \text{Beta}(k, n + 1 - k). \tag{S4}$$

This is Vovk's training-conditional result.[2,3] Under infinite exchangeability the same Beta law holds conditional on the de Finetti directing distribution, and therefore also marginally (Section S3).

**Correction.** For a request $(\alpha, \delta)$ and batch size $m$, let $x^\star = \lceil (1-\alpha)m \rceil$, computed in integer arithmetic. The correction selects the least conservative order statistic, that is the largest $u$, such that

$$\Pr\left(X \geq x^\star\right) \geq 1 - \delta, \qquad X \sim \text{BetaBinomial}(m; n + 1 - u, u). \tag{S5}$$

The batch-free version replaces the Beta–Binomial by $\text{Beta}(n+1-u, u)$ and the event by $\{Z \geq 1 - \alpha\}$. The procedure is applied separately to each class; for class $y$, $m$ is the number of future compounds of that class, and the guarantee holds conditional on the class labels.

**Algorithm.** (i) Set $x^\star = \lceil (1-\alpha)m \rceil$. (ii) For $u = 1, \ldots, n$ compute the upper-tail probability above. (iii) Return the largest $u$ for which it is at least $1 - \delta$, together with $k = n + 1 - u$ and the deployed level $u/(n+1)$; if there is none, return *infeasible*.

**Feasibility.** The most conservative choice $u = 1$ gives coverage distributed as $\text{Beta}(n, 1)$. A batch-free request is therefore feasible if and only if

$$\alpha \geq 1 - \delta^{1/n}. \tag{S6}$$

A batch request is feasible if and only if $\Pr(X \geq x^\star) \geq 1 - \delta$ for $X \sim \text{BetaBinomial}(m; n, 1)$. Distinct requests often select the same $u$ and hence the same deployed model (Fig. S1).

**Baselines.** The standard choice is $k = \lceil (n+1)(1-\alpha) \rceil$. The DKW-based correction replaces $\alpha$ by $\alpha - \sqrt{\log(1/\delta)/(2n)}$ (one-sided bound with the Massart constant),[4] and falls back to $k = n$ when this value is not positive.

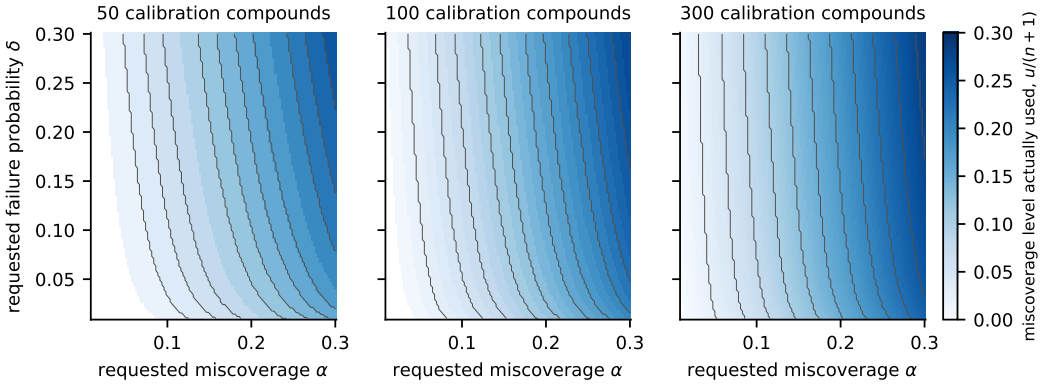



**Figure S1:** Deployed miscoverage level selected by the batch-free correction as a function of the request $(\alpha, \delta)$, for three calibration sizes. Contours mark deployed levels in steps of 0.025; white: infeasible requests ($\alpha < 1 - \delta^{1/n}$).

# S3 Prediction intervals for future-batch counts from an audit set

**Construction.** Fix the calibration set, hence the thresholds and the prediction-set map. For a fixed indicator $g$, let $I_1, \ldots, I_n$ be its values on the test compounds and $I_{n+1}, \ldots, I_{n+m}$ on the future batch. The audit and future compounds are exchangeable with one another given the calibration set, which is what this construction needs and all that it needs: the calibration set enters only through the thresholds it fixed, so the requirement is conditional on it and it need not be exchangeable with the audit or the batch. Nor need it be independent of them: a random division of a fixed library makes the three parts dependent through the library's finite size, and the construction is valid there, as the fixed-library simulation below confirms. Given the calibration set the $n+m$ indicators are therefore exchangeable. Let $K = \sum_{i \le n} I_i$, $K_m = \sum_{i > n} I_i$ and $T = K + K_m$. Conditional on $T = t$, every placement of the $t$ ones among the $n+m$ positions is equally likely, so

$$K \mid T = t \;\sim\; \text{Hypergeometric}(n+m, t, n), \tag{S7}$$

whatever the distribution of the data. For each candidate $j$ for $K_m$, test $T = K + j$ with this distribution and keep $j$ if neither one-sided test rejects at level $\gamma/2$. The accepted values form an interval $[L, U]$ with $\Pr(L \le K_m \le U) \ge 1 - \gamma$.

**Closed form.** For integers $0 \le k \le n$ and $0 \le j \le m$,

$$\Pr\big(K \ge k \mid T = k + j\big) = \Pr\big(B \le j\big), \qquad B \sim \text{BetaBinomial}(m; k, n-k+1), \tag{S8}$$

and symmetrically for the other tail with parameters $(k+1, n-k)$.[5] Hence

$$L = \min\{j : F_{\text{BB}(m;k,n-k+1)}(j) > \gamma/2\}, \qquad U = \text{the } 1-\gamma/2 \text{ quantile of BetaBinomial}(m; k+1, n-k), \tag{S9}$$

with $L = 0$ if $k = 0$ and $U = m$ if $k = n$. We verified both identities numerically for $n \in \{20, 57, 200\}$, $m \in \{10, 100, 1000\}$ and all $k$ and $j$; the largest discrepancy was $6 \times 10^{-13}$ (script in Section S16). The interval $[L, U]$ reaches at least $1 - \gamma$ even conditional on each total $T$. It contains the Bayesian posterior predictive intervals with priors $\text{Beta}(1,1)$ and $\text{Beta}(1/2, 1/2)$, so their coverage is never higher than that of the exact interval.

**Sizing the audit set.** For planning, the exact limits are close to a normal approximation: for an outcome of frequency $p$, $n$ audit compounds and a batch of $m$, the 95% interval for the future count is about $mp \pm 2m\sqrt{p(1-p)(1/n + 1/m)}$. The term $1/n$ is the uncertainty of the audited frequency and $1/m$ the variation of a finite batch; the second does not vanish however large the audit. For AqSolDB setting A ($p = 0.23$ false leads, $n = 384$, $m = 1000$) the approximation gives $232 \pm 51$, against the exact $[183, 285]$. Joint bounds over $J$ settings and $C$ outcomes replace the factor 2 by the corresponding normal quantile, about 3.8 for $JC = 576$ and $\varepsilon = 0.05$.

Inverting the approximation sizes the audit. Letting the audit grow without limit leaves the half-width at $h_\infty = 2\sqrt{m\,p(1-p)}$, which is the narrowest interval any audit can buy for a batch of $m$, and writing the target half-width as $r\,h_\infty$ for some $r > 1$ gives

$$n = \frac{m}{r^2 - 1}. \tag{S10}$$

The rate and the batch size cancel, so where a design sits above its floor is fixed by $n/m$ alone: $n = m/3$ gives twice the floor, $n = m$ a factor $\sqrt{2}$, $n = 2.3m$ a factor 1.2, $n = 4.8m$ a factor 1.1 and $n \approx 10m$ a factor 1.05. Checked against the exact limits at $p$ from 0.02 to 0.50 and $m$ from 96 to 1000, the realized ratio matches $r$ to within a few percent, the discrepancies being those of discreteness at small expected counts. The floor does depend on the rate, and relative to the expected count $mp$ it is $2\sqrt{(1-p)/(pm)}$, which is 44% at $p = 0.02$ and $m = 1000$ against 9% at $p = 0.5$ and $m = 500$: at the rates of a primary screen the batch term dominates, and a larger audit cannot repair it.

**Table S1:** Fraction of calibration draws in which coverage over a batch of $m = 100$ compounds fell below 90% ($\alpha = \delta = 0.10$; absolute Cauchy scores; $10^6$ draws). $u$: selected index; deployed level $u/(n+1)$. Theory (batch): Beta–Binomial value. Theory (batch-free): $\Pr(\text{coverage} < 0.9)$ under the Beta law. DKW rows use the one-sided constant $\log(1/\delta)$. All rules were evaluated on the same draws. The batch-free correction selects the same $u$ as the batch correction for $n \le 100$, and $u = 10, 15, 19, 23, 41$ for $n = 150, 200, 250, 300, 500$, with observed batch violation fractions 0.118, 0.168, 0.170, 0.171 and 0.212.

| $n$ | Threshold | $u$ | level | Observed | Theory (batch) | Theory (batch-free) |
|---|---|---|---|---|---|---|
| 50 | standard | 5 | 0.098 | 0.397 | 0.396 | 0.431 |
| | correction | 2 | 0.039 | 0.047 | 0.047 | 0.034 |
| | DKW (one-sided) | 1 | 0.020 | 0.010 | 0.010 | 0.005 |
| 75 | standard | 7 | 0.092 | 0.347 | 0.346 | 0.367 |
| | correction | 4 | 0.053 | 0.077 | 0.077 | 0.050 |
| | DKW (one-sided) | 1 | 0.013 | 0.002 | 0.002 | 0.000 |
| 100 | standard | 10 | 0.099 | 0.408 | 0.407 | 0.451 |
| | correction | 6 | 0.059 | 0.096 | 0.096 | 0.058 |
| | DKW (one-sided) | 1 | 0.010 | 0.000 | 0.000 | 0.000 |
| 150 | standard | 15 | 0.099 | 0.411 | 0.411 | 0.460 |
| | correction | 9 | 0.060 | 0.080 | 0.080 | 0.031 |
| | DKW (one-sided) | 1 | 0.007 | 0.000 | 0.000 | 0.000 |
| 200 | standard | 20 | 0.100 | 0.413 | 0.412 | 0.466 |
| | correction | 13 | 0.065 | 0.099 | 0.098 | 0.032 |
| | DKW (one-sided) | 4 | 0.020 | 0.001 | 0.001 | 0.000 |
| 250 | standard | 25 | 0.100 | 0.413 | 0.413 | 0.469 |
| | correction | 16 | 0.064 | 0.086 | 0.086 | 0.018 |
| | DKW (one-sided) | 8 | 0.032 | 0.003 | 0.003 | 0.000 |
| 300 | standard | 30 | 0.100 | 0.414 | 0.414 | 0.472 |
| | correction | 20 | 0.066 | 0.097 | 0.097 | 0.017 |
| | DKW (one-sided) | 11 | 0.036 | 0.005 | 0.005 | 0.000 |
| 500 | standard | 50 | 0.100 | 0.415 | 0.415 | 0.478 |
| | correction | 34 | 0.068 | 0.095 | 0.095 | 0.005 |
| | DKW (one-sided) | 26 | 0.052 | 0.024 | 0.025 | 0.000 |

**Why the exact limits are used.** The decomposition above is the ordinary large-sample one. Treating the candidates as independent draws at rate $p$ and the audited frequency as $\hat{p} = K_n/n$, the prediction error of $m\hat{p}$ has variance $mp(1-p) + m^2 p(1-p)/n$, which is $m^2 p(1-p)(1/n + 1/m)$: the first term is the batch and the second the estimate. Equation S10 is exact in that limit rather than an approximation to it, since the ratio of half-widths is $\sqrt{1+m/n}$ by construction. None of this is new, and the exact construction we use is its finite-population counterpart — predicting the defectives in the remainder of a lot from a sample of it[5–7] — so the two are descriptions of one quantity.

They do not agree where screens operate, which is why the exact limits are used throughout. The sizing figures quoted in this paragraph and in the main text invert a *one-sided* lower limit, placing the whole of $\gamma = 0.05$ in the lower tail rather than $\gamma/2$ in each tail as the two-sided limits above do; the one-sided form is the right object for a question about a shortfall only. The two conventions differ by a factor that grows as the audit shrinks — about 1.1 at an audit of 1000, 1.2 at 384, and 2.0 at 49 — so the convention must be read with the figure. Because $\gamma$ denotes a one-sided level in the sizing figures and a two-sided one elsewhere, we write $\gamma_1$ for the former where both appear. Sizing a round for 20 hits, the large-sample rule and the exact limits agree to within 3% whenever the audit has seen about twenty in-class compounds or more: at $p = 0.5$ for every audit size, at $p = 0.10$ from $n = 384$ (38 seen) and at $p = 0.02$ from $n = 1000$ (20 seen). Below that the large-sample rule is conservative and then wrong. At $p = 0.02$ with $n = 384$, 8 in-class compounds seen, it asks for 2690 candidates against the exact 2056, and at $p = 0.10$ with $n = 49$, 5 seen, 736 against 510. At $p = 0.02$ with $n = 49$, one in-class compound seen, it returns no finite batch at all: the lower bound it places on the rate is not positive, so no batch size makes the bound reach the target. The rule needs $n > z^2(1-p)/p$ audit candidates before it will admit any finite answer, about 133 at a rate of 2%, whereas the exact limits return 19 091. A team following the approximation at a primary screen's hit rate would therefore over-size its rounds, and on a small audit would abandon a plan that is in fact feasible.

## S3.1 Composing a rate bound with a binomial tail

A natural objection to the claim that a rate bound does not price a round is that one can always compose: bound the rate from the audit, then take a binomial upper quantile at that rate, and union the two failure events. The composition is valid and it is the right thing to compare against, so we price it rather than dismiss it.

Both sides are one-sided upper bounds at a total level of 0.05. The composition is given 0.025 at each stage, since a split is what makes the union valid at 0.05 overall, and the rate stage is given the most generous instrument available:

an exact Clopper–Pearson upper limit, not the Hoeffding–Bentkus or betting bounds that risk-controlling prediction sets actually use, which are looser at these audit sizes. The exact inversion places the whole of 0.05 in its upper tail. Over the seven configurations this paper reports the composed limit is 1.11 to 1.60 times the exact one. Expressed as headroom above the expected count — the quantity a planner reserves, and the one in which the two differ most — the worst case is a plate of 96 priced from an audit of 384 at a 2% rate, where the expected count is 2.0, the exact limit is 5 and the composed limit 8, so the reserve doubles from three compounds to six.

Two things beyond width separate them. The composition pays for the rate and then for the tail, where conditioning on the audit-and-batch total pays once. And its second stage treats the batch count as binomial given the rate, which holds when the batch is an independent sample from a fixed distribution; a finite exchangeable sequence need not be a mixture of independent ones,[8] and the hypergeometric argument of Section 2.5 needs no such representation. The comparison is reproduced by the composition pricing described in the data availability statement.

**Target of the interval.** The interval covers the realized future count $K_m$; no fixed rate is assumed. If the compound source is regarded as infinitely exchangeable, de Finetti's theorem[9] gives a random distribution $Q$ such that, conditional on $Q$, compounds are independent draws from $Q$. Conditional on $Q$ and the calibration set the test indicators are then independent Bernoulli variables with success probability

$$p_g(Q) = \Pr_Q\big(g(Y, C(X)) = 1\big), \tag{S11}$$

so $K$ is Binomial and the Clopper–Pearson interval covers $p_g(Q)$ with probability at least $1-\gamma$, conditionally and therefore marginally. $p_g(Q)$ is the long-run frequency of the outcome in this deployment. The probability $\Pr(g = 1 \mid \mathscr{D}_{\text{cal}})$, averaged over $Q$, is not the success probability of independent trials, and intervals for it are not valid in general; in the two-stage simulation of Table S7 Clopper–Pearson coverage of that quantity fell to 73%. Finite exchangeable sequences need not be mixtures of independent sequences;[8] the hypergeometric construction above does not need this representation.

**Several outcomes.** Each interval is valid for one indicator fixed before seeing the audit set. For $J$ indicators, intervals at level $\gamma/J$ hold simultaneously. A joint region for the full outcome table can be obtained from the multivariate hypergeometric distribution of the cell counts given their totals.

**Re-estimating between rounds.** A campaign that screens a first batch, revises the rate and sizes the next has looked at the data more than once, which a fixed-sample interval does not cover; dividing the confidence level across a declared number of looks restores a simultaneous statement over those looks. Three looks rather than one raise the required batch from 309 to 350 candidates at a rate of 10%, and from 2056 to 2605 at 2%, so declaring three looks in advance costs between a seventh and a quarter of a batch, 27% at the lower rate. The conditions are that the number of looks is fixed before the campaign starts, and that the correction is over the looks only. Stopping when the hits so far look sufficient is not covered: the continuation is then data-dependent, and an interval reported at the stopping look is conditioned on having reached it. A campaign that wants to stop when it likes needs an anytime-valid construction — an alpha-spending boundary or a confidence sequence — rather than a division of the level,[10–12] which we do not develop here. Nor does any of this repair the selection in the inversion itself, discussed above; it addresses only the multiplicity of looking more than once. Which data the revised rate comes from also matters, and if it comes from a batch the model selected, the revision inherits that selection.

# S4 Prediction intervals from the calibration set alone

**Events.** Use the notation of Section S6: $u = \hat{p}(1 \mid x)$, $0 \in C(x)$ iff $u \le \tau_0$, and $1 \in C(x)$ iff $u \ge 1-\tau_1$. Here $\tau_0$ is the $k_0$-th smallest of the $n_0$ out-class calibration scores $u$, and $\tau_1$ is the $k_1$-th smallest of the $n_1$ in-class calibration scores $1-u$. For out-class compounds define

$$A_0 = \{u \le \tau_0\} \text{ (coverage)}, \qquad B_0 = \{u \ge 1-\tau_1\} \text{ (labelled in-class)}, \tag{S12}$$

and for in-class compounds $A_1 = \{u \ge 1-\tau_1\}$ and $B_1 = \{u \le \tau_0\}$.

**Own-class events.** Let $m_y$ be the number of future compounds of class $y$. The number of them in $A_y$ is the number of $m_y$ exchangeable scores at or below the $k_y$-th of $n_y$ calibration scores. Its distribution is exactly[1]

$$\text{BetaBinomial}(m_y;\, k_y,\, n_y + 1 - k_y), \tag{S13}$$

and its equal-tailed quantiles give an interval.

**Cross-class events.** $B_0$ depends on the data only through $\tau_1$, which is a function of the in-class calibration compounds. Given the labels and the in-class calibration compounds, the out-class calibration compounds and the future out-class compounds are exchangeable. Let $c_0$ be the number of out-class calibration compounds in $B_0$. The audit-set construction of Section S3, with $n = n_0$, $K = c_0$ and $m = m_0$, then gives an interval for the number of future out-class compounds in $B_0$. The same holds for $B_1$ with the roles of the classes exchanged.

**Cells.** By Proposition S1, if $\tau_0 + \tau_1 \geq 1$ then $A_0^c \subseteq B_0$, and the out-class row is

$$\{1\} = A_0^c, \qquad \{0\} = B_0^c, \qquad \text{both} = A_0 \cap B_0, \qquad \varnothing = \emptyset. \tag{S14}$$

If $\tau_0 + \tau_1 < 1$ then $A_0 \cap B_0 = \emptyset$, and

$$\{0\} = A_0, \qquad \{1\} = B_0, \qquad \varnothing = A_0^c \cap B_0^c, \qquad \text{both} = \emptyset. \tag{S15}$$

The regime is known once the thresholds are set, but it depends on the calibration data of both classes, so a cell's identity with one event or the other is data dependent. We therefore compute the interval for $|A_0|$ (rank law) and the interval for $|B_0|$ (cross-class) each at level $1 - \gamma/2$. Both then hold simultaneously with probability at least $1 - \gamma$, whatever the regime. Every cell of the row is bounded by arithmetic on the two: in the two-label regime, $\{1\}$ and $\{0\}$ are bounded by the complements of $A_0$ and $B_0$, and the both-labels cell by $|A_0| + |B_0| - m_0$; in the empty regime, $\{0\}$ and $\{1\}$ are bounded by $A_0$ and $B_0$, and the empty cell by $m_0 - |A_0| - |B_0|$. The intervals for all cells of the row hold jointly at level $1 - \gamma$. The in-class row is analogous: missed in-class compounds are $A_1^c$ (rank law) and confident hits are $B_1^c$ (cross-class) in the two-label regime.

**Mixed batches.** The guarantees above are conditional on $m_y$. For a batch of $m$ compounds of unknown composition, the number of in-class compounds $m_1$ has the audit-set interval from the calibration class counts ($K = n_1$ out of $n_0 + n_1$). We compute the interval for $m_1$ at level $1 - \gamma/2$, and each of the four event intervals $(A_0, B_0, A_1, B_1)$ at level $1 - \gamma/8$, for every $m_1$ in that interval, and take the union over $m_1$. By the union bound the five statements hold together with probability at least $1 - \gamma$, so the intervals for all cells of the mixed batch hold jointly at level $1 - \gamma$. The union over $m_1$ is conservative; mixing the class-wise laws over the exact distribution of $m_1$ would give shorter intervals.

**When it does not apply.** If scores are not class probabilities ($s(x,0) + s(x,1) \neq 1$), the sets $A_0 \cap B_0$ and $A_0^c \cap B_0^c$ can both have positive probability. Their counts then depend on the joint behaviour of two thresholds on the same compounds, and an audit set is needed. Evaluating a class's own threshold on the compounds that set it, beyond the rank law, is not allowed (Section S5).

**Validation.** Table S2 reports coverage and width from the simulation of Section S8.

**Table S2:** Calibration-only intervals (nominal 95%), future batch of $m = 1000$. Class-conditional: coverage of the row intervals given the number of future compounds of the class. Mixed: coverage for a batch of unknown composition (union over the interval for the number of in-class; levels as in Section S4). Widths are for the mixed batch. Split budget: the same $n$ labelled compounds split equally into calibration and test sets, with test-set intervals. $n = 300$: 2897 replicates (103 skipped with fewer than 20 calibration compounds in a class), all mixed-batch cells covered together in 99.7%; $n = 1000$: 3000 replicates (0 skipped with fewer than 20 calibration compounds in a class), all mixed-batch cells covered together in 99.5%.

| $n$ | Cell | Coverage, calibration only | | Mean width | | Coverage |
|---|---|---|---|---|---|---|
| | | class-conditional | mixed | calibration only | split budget | split budget |
| 300 | out-class $\{0\}$ (correct rejection) | 0.978 | 1.000 | 202.9 | 158.7 | 0.964 |
| | out-class $\{1\}$ (false lead) | 0.977 | 0.999 | 78.9 | 69.4 | 0.972 |
| | out-class deferred | 0.999 | 1.000 | 180.7 | 106.2 | 0.973 |
| | in-class $\{1\}$ (confident hit) | 0.981 | 0.999 | 164.8 | 138.8 | 0.965 |
| | in-class $\{0\}$ (missed in-class) | 0.989 | 0.999 | 47.8 | 42.1 | 0.988 |
| | in-class deferred | 0.998 | 1.000 | 104.8 | 68.8 | 0.982 |
| 1000 | out-class $\{0\}$ (correct rejection) | 0.979 | 1.000 | 138.9 | 99.7 | 0.960 |
| | out-class $\{1\}$ (false lead) | 0.984 | 0.999 | 52.9 | 43.7 | 0.972 |
| | out-class deferred | 1.000 | 1.000 | 123.7 | 59.8 | 0.969 |
| | in-class $\{1\}$ (confident hit) | 0.981 | 0.998 | 111.2 | 85.2 | 0.946 |
| | in-class $\{0\}$ (missed in-class) | 0.987 | 0.999 | 31.8 | 24.8 | 0.977 |
| | in-class deferred | 1.000 | 1.000 | 71.3 | 36.9 | 0.976 |

## S5 Same-class reuse of the calibration set, and the leave-one-out estimate

Let $I_i(t) = \mathbf{1}\{S_i \le t\}$ for the $n$ calibration scores of one class. If $\tau = S_{(k)} = t$, then exactly $k$ indicators equal one, so the indicators are not independent given the threshold:

$$\mathrm{Cov}\big(I_i(t), I_j(t) \mid \tau = t\big) = -\frac{k(n-k)}{n^2(n-1)} < 0, \qquad i \neq j,\ k < n. \tag{S16}$$

Outcome counts that evaluate a class's own threshold on the calibration compounds of that class inherit this dependence and are not exchangeable with future counts; only the rank law of Section S2 describes them exactly. Cross-class events are not affected (Section S4). Intervals built from them are too narrow, and their centres can also be biased near thresholds.

**Leave-one-out estimate.** When no audit set is available, each calibration compound $i$ can be assigned to a prediction set using thresholds recomputed without it. The pooled cell counts give a leave-one-out estimate of the outcome table.[13,14] The thresholds change from one compound to the next, so this estimate carries no guarantee. With class probabilities, the calibration-only intervals of Section S4 make it unnecessary; for other scores it is useful only for exploring settings before an audit set is collected.

## S6 Binary prediction sets with probability scores

Write $u = s(x,0) = \hat{p}(1 \mid x)$. If $\hat{p}(0 \mid x) + \hat{p}(1 \mid x) = 1$ then $s(x,1) = 1 - u$, and the sets are determined by $u$ alone: $0 \in C(x)$ iff $u \le \tau_0$ and $1 \in C(x)$ iff $u \ge 1 - \tau_1$.

**Proposition S1.** *With probability scores, (i) the two-label set has positive probability only if $\tau_0 + \tau_1 > 1$, in which case $u \in [1-\tau_1, \tau_0]$; (ii) the empty set has positive probability only if $\tau_0 + \tau_1 < 1$, in which case $u \in (\tau_0, 1-\tau_1)$; (iii) if $\tau_0 + \tau_1 = 1$ (and $u$ has a continuous distribution) only single-label sets occur.*

*Proof.* Direct from the two conditions on $u$. □

When $\tau_0 + \tau_1 > 1$, compounds with $u < 1 - \tau_1$ receive $\{0\}$, those with $u > \tau_0$ receive $\{1\}$, and those in between receive both labels (main text Fig. 1). Raising $\tau_1$ moves compounds from $\{0\}$ to both labels; raising $\tau_0$ moves compounds from $\{1\}$ to both labels. Each single-label set is thus controlled by the threshold of the other class. This is why changing the request for one class can change the outcomes of the other.

## S7 Cost ratios for which acting on single labels is cheapest

**Table S3:** Expected cost relative to Chow's optimal rule (%), constrained (coverage $\ge 0.90$ for both classes) / default ($\alpha = 0.10$ for both classes). $\lambda$: cost of a miss; $\rho$: cost of an assay; both relative to a false lead.

| Scores | $\lambda$ | $\rho = 0.1$ | $\rho = 0.3$ | $\rho = 0.5$ | $\rho = 0.8$ |
|---|---|---|---|---|---|
| well separated | 2 | 0 / 35 | 0 / 38 | 0 / 68 | 0 / 138 |
| | 5 | 0 / 23 | 0 / 12 | 0 / 25 | 0 / 58 |
| | 10 | 0 / 27 | 0 / 2 | 0 / 7 | 0 / 26 |
| | 20 | 0 / 48 | 0 / 3 | 0 / 0 | 0 / 9 |
| | 50 | 0 / 119 | 0 / 26 | 0 / 9 | 0 / 5 |
| overlapping | 2 | 0 / 134 | 4 / 41 | 18 / 31 | 47 / 47 |
| | 5 | 0 / 152 | 0 / 31 | 0 / 9 | 6 / 5 |
| | 10 | 0 / 220 | 0 / 49 | 0 / 15 | 0 / 1 |
| | 20 | 0 / 372 | 0 / 100 | 0 / 44 | 0 / 16 |
| | 50 | 0 / 838 | 0 / 262 | 0 / 144 | 0 / 80 |
| balanced, overlapping | 2 | 0 / 132 | 0 / 21 | 1 / 5 | 14 / 14 |
| | 5 | 0 / 319 | 0 / 91 | 1 / 48 | 12 / 35 |
| | 10 | 0 / 632 | 0 / 212 | 1 / 127 | 10 / 89 |
| | 20 | 0 / 1259 | 0 / 453 | 1 / 287 | 9 / 203 |
| | 50 | 0 / 3133 | 0 / 1171 | 0 / 762 | 8 / 543 |

Consider actions: label 0, label 1, or defer, with costs $c_{\text{miss}}$ for assigning 0 to a compound of class 1, $c_{\text{lead}}$ for assigning 1 to a compound of class 0, $c_{\text{defer}}$ for deferring, and zero for correct labels. For a prediction set $c$ with

positive probability let

$$\eta_c = \Pr\big(Y = 1 \mid C(X) = c\big) = \frac{p_{1,c}}{p_{0,c} + p_{1,c}}. \tag{S17}$$

A rule that acts only on the prediction set has expected costs $\eta_c c_{\text{miss}}$ (label 0), $(1-\eta_c)c_{\text{lead}}$ (label 1) and $c_{\text{defer}}$ (defer). With $\lambda = c_{\text{miss}}/c_{\text{lead}}$ and $\rho = c_{\text{defer}}/c_{\text{lead}}$, the rule "$\{1\} \to$ label 1, $\{0\} \to$ label 0, otherwise defer" minimizes expected cost if and only if all of the following hold for the sets that occur: [15]

$$\eta_{\{1\}} \geq \max\Big\{\frac{1}{1+\lambda},\, 1-\rho\Big\}, \tag{S18}$$

$$\eta_{\{0\}} \leq \min\Big\{\frac{1}{1+\lambda},\, \frac{\rho}{\lambda}\Big\}, \tag{S19}$$

$$\frac{\rho}{\lambda} \leq \eta_c \leq 1-\rho \quad \text{for } c \in \{\{0,1\}, \varnothing\}. \tag{S20}$$

Each condition is linear in $(\lambda, \rho)$, so the admissible region is a polygon, possibly empty. Deferral can be optimal only if $c_{\text{defer}} \leq c_{\text{miss}} c_{\text{lead}}/(c_{\text{miss}} + c_{\text{lead}})$. The $\eta_c$ are estimated from the outcome table.

**Set-based rules contain Chow's rule.** Let $\eta(u) = \Pr(Y = 1 \mid u)$, and suppose $\eta$ is nondecreasing in $u$, which is an assumption about the model and not a consequence of its ranking well; Table S9 shows it failing on nine of twelve endpoints. In units of $c_{\text{lead}}$, the expected costs of the three actions for a compound with score $u$ are $\lambda\eta(u)$ (label 0), $1-\eta(u)$ (label 1) and $\rho$ (defer). Chow's optimal rule [15] takes the cheapest action at each $u$. Because $\eta$ is monotone, the optimal rule labels 0 for $u < l$, defers for $l \leq u \leq h$ and labels 1 for $u > h$, for some $l \leq h$. The set-based rule with $\tau_0 = h$ and $\tau_1 = 1 - l$ produces exactly this partition (Section S6). If deferral is never optimal, then $l = h$ and the thresholds satisfy $\tau_0 + \tau_1 = 1$. Conformal thresholds are calibration order statistics, so $(l, h)$ can be matched only up to the spacing of the calibration scores. In addition, a sweep over a finite grid of requests reaches only the threshold pairs on that grid. Choosing the request by cost therefore loses nothing relative to Chow's rule beyond the resolution of the calibration scores and of the request grid, *provided* $\eta$ is monotone. Requiring coverage of at least $1-\alpha$ for both classes restricts $(l, h)$, and its cost can be computed (Table S3). The argument needs $\eta$ to be monotone, not calibrated: the set-based rule uses only the ordering of $u$. If $\eta$ is not monotone, Chow's rule on calibrated probabilities is not a two-threshold rule on $u$ and can be cheaper; Table S9 gives a monotonicity diagnostic for the Tox21 models. The containment is a statement about population rules. On finite data, a set rule chosen from calibration estimates and Chow's rule on fitted probabilities are both estimates, and either can be cheaper (Table S13). What the set rule adds is the interval guarantee for its outcome counts, which after selection holds only with the correction of the Selection paragraph below.

**Uncertainty in $\eta_c$.** The conditions above use estimated within-set fractions $\hat{\eta}_c$. A robust region is the set of $(\lambda, \rho)$ for which the conditions hold for every $\eta_c$ in simultaneous Clopper–Pearson intervals: Bonferroni over the sets that occur, computed from the counts $p_{0,c}, p_{1,c}$ of the outcome table. The conditions are monotone in each $\eta_c$, so this needs only the interval endpoints. These intervals need the joint counts of label and prediction set on compounds not used to set the thresholds, that is a labelled audit set; the calibration-only intervals of Section S4 bound the cells of each row separately and do not by themselves give intervals for the within-set fractions. In the solubility example the robust region therefore uses the audit set.

**Selection.** A sweep over a grid of $J$ requests yields at most $J$ distinct rules. If intervals for all rules are computed at level $\gamma/J$, they hold simultaneously with probability at least $1-\gamma$, so the intervals of any rule chosen by looking at them remain valid. Choosing by an estimated cost or by a capacity constraint is then safe. We divide by the size of the request grid, which is fixed in advance; dividing by the number of distinct rules is also valid when the thresholds are set from data independent of the data used for the intervals (as in the audit-set design), because the set of distinct rules is then fixed given the calibration set. Bonferroni is conservative when $J$ is large. Sequential procedures such as fixed-sequence testing [16] control error for hypotheses about fixed parameters, for example the long-run deferral rate $p_g(Q)$ of Section S3; they do not by themselves protect statements about realized future counts, because which settings exceed a capacity in the future batch is itself random. We therefore use Bonferroni for all statements about future counts.

**The usual set-to-action rule is max–min optimal.** Measure costs in units of a false lead: a miss costs $\lambda$, a deferral $\rho$, a correct call 0. Kiyani *et al.* show that a decision maker who knows only a prediction set with marginal coverage, and wants a high-probability guarantee on its utility, should take the action with the best worst case over the labels in the set. [17] For $\{1\}$ the worst-case cost of labelling 1 is 0, so the rule labels 1; likewise for $\{0\}$. For $\{0,1\}$ the worst cases are $\lambda$ (label 0), 1 (label 1) and $\rho$ (defer), so the rule defers whenever $\rho \leq \min(\lambda, 1)$. Empty sets occur here by design rather than only on miscoverage; we defer them. Acting on single labels and deferring the rest is therefore the

max–min action for every cost with $\rho \leq \min(\lambda, 1)$, whatever the coverage setting. Expected-cost (Chow) optimality, the criterion that governs the average over a campaign, holds only where the within-set fractions satisfy the conditions above; the two criteria answer different questions. Kiyani *et al.* also show that, for a given utility, the optimal prediction sets are level sets of utility rather than of a score, so class-conditional threshold sets are not in general the optimal sets.

**Scenario maps (main text Fig. 3, Fig. S3).** For one split, each request on the $12 \times 12$ grid ($\delta = 0.10$, batch correction with planned class counts) sets thresholds on the calibration set. For a future batch of $M = 1000$, the audit set gives the point estimate of each outcome count and a one-sided bound from the construction of Section S3: upper bounds for deferrals (measurements), false leads and missed in-class compounds, a lower bound for confirmed in-class compounds. Each bound is computed at level $\varepsilon/(JC)$ with $\varepsilon = 0.05$, $J = 144$ requests and $C = 4$ bounded outcomes ($8.7 \times 10^{-5}$), so that all bounds of all requests hold together with probability at least 95%. A scenario lists limits on some of these bounds; a request is feasible if all its bounds satisfy the limits, and the feasible request with the fewest expected false leads is chosen. Because every bound of every request holds together, the chosen setting satisfies its constraints with probability at least 95% although it was chosen by looking at the audit set. The CALiSol-23 maps use the first random split of the electrolyte screen. If the objective and the limits are monotone in the outcomes, an optimal feasible setting lies on the Pareto front of the vectors (upper bounds on deferrals, false leads and missed in-class compounds; lower bound on confirmed in-class compounds; and, for the point objective, expected false leads): if setting $b$ dominates setting $a$, then $b$ meets every limit $a$ meets and is at least as good under the objective. On these maps the front held 105 of the 144 distinct settings for AqSolDB and, for CALiSol-23, whose 144 requests give 132 distinct settings but only 107 distinct audited outcome vectors, 33 of those 107, which is 41 of the 144 requests, and the chosen setting lay on it in all four scenarios; adding the expected false leads to the vector did not change the front. For the electrolyte screen (Fig. S3), with at most 200 measurements per 1000, at least 60 confirmed good electrolytes and at most 40 missed, 56 of 144 requests were feasible; halving the measurement budget left 19 and raised the expected false leads of the best setting from 53 to 67 per 1000. Because deferrals arise from both-label sets for some requests and from empty sets for others, the measurement constraint bounds the feasible region from two sides. For AqSolDB, 23–48 settings per split were expected-cost optimal for some cost ratio on the grid of Fig. S2, and the ratios for which acting on the sets stays optimal over the uncertainty of the within-set fractions covered 0.6–6.1% of the grid; the choice by constraints in the main text needs no cost ratio.

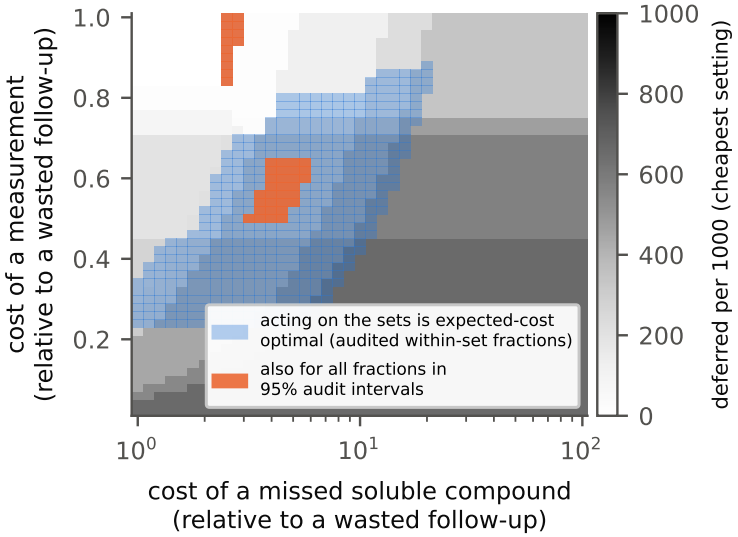



**Figure S2:** AqSolDB lipophilic screen (split 0): for each pair of cost ratios, the number of deferrals of the setting with the lowest expected cost (grey), computed from audited counts. Blue: cost ratios for which acting on single labels and deferring the rest is expected-cost optimal for that setting at the audited within-set fractions. Orange: where this holds for all fractions within simultaneous 95% audit intervals. Regions are evaluated cell by cell.

# S8 Simulation protocols and additional results

**Threshold correction (Table S1). Coverage against audit size for every outcome is shown in Fig. S5. The scaffold-shift simulation is shown in Fig. S4.** Calibration scores were drawn from the absolute value of a Cauchy distribution,

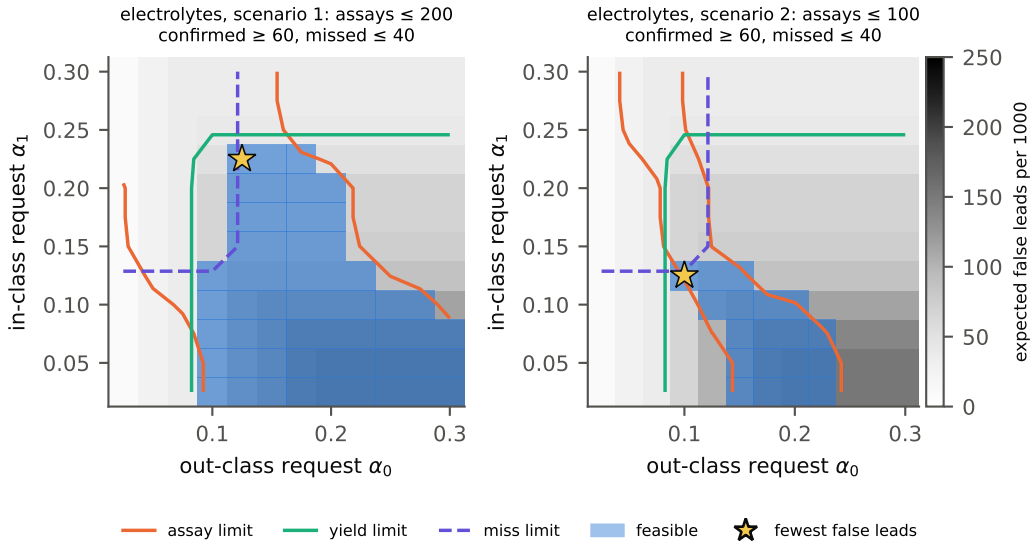



**Figure S3:** Scenario maps for the electrolyte screen (CALiSol-23, first random split), as in main text Fig. 3. Left: at most 200 measurements, at least 60 confirmed good electrolytes and at most 40 missed per 1000. Right: at most 100 measurements. The measurement limit bounds the feasible region on two sides because deferrals come from both-label sets for some requests and from empty sets for others.

and the future batch ($m = 100$) from the same distribution, for $10^6$ independent calibration draws per $n$; all threshold rules were evaluated on the same draws.

**Prediction intervals (main text Fig. S5).** Each replicate drew a data distribution $Q$: prevalence $\pi \sim \mathrm{U}(0.05, 0.5)$; predicted probability of being in-class $\mathrm{Beta}(a_1, b_1)$ for in-class compounds with $a_1 \sim \mathrm{U}(2, 10)$, $b_1 \sim \mathrm{U}(2, 4)$; and $\mathrm{Beta}(a_0, b_0)$ for out-class compounds with $a_0 \sim \mathrm{U}(1.5, 3)$, $b_0 \sim \mathrm{U}(5, 9)$. It then drew 1000 calibration compounds, $n$ test compounds and $m = 1000$ future compounds independently from $Q$. Over replicates the data are exchangeable but not independent. Thresholds used the batch-free correction with $\alpha = \delta = 0.10$ for each class. Replicates with fewer than 30 calibration compounds in a class were skipped (none occurred). The long-run cell probabilities given $Q$ and the thresholds were computed exactly from the Beta distribution functions. Besides the audit interval, three common alternatives were evaluated: the Bayesian posterior predictive with a uniform prior, Beta–Binomial$(m; K+1, n-K+1)$; the plug-in interval, Binomial$(m, K/n)$ quantiles; and $m$ times the Clopper–Pearson interval for the rate, rounded outward. Results are in Table S6. Because the data are independent given $Q$, this design does not separate validity under exchangeability from validity under independence. The future count varies for two reasons: the audited rate is uncertain, which adds a variance of about $m^2 p(1-p)/n$, and the batch itself is a sample, which adds about $mp(1-p)$. The plug-in interval keeps only the second term and is too narrow by a factor of about $\sqrt{1+m/n}$; $m$ times the Clopper–Pearson interval keeps only the first and is too narrow by about $\sqrt{1+n/m}$. Each fails where the term it omits dominates: the plug-in for small audits, the scaled Clopper–Pearson interval for audits as large as the batch, and for any audit when the batch is small. The audit interval accounts for both terms. Its limits use the same Beta parameters as the Clopper–Pearson limits but take Beta–Binomial rather than Beta quantiles, and as $m \to \infty$, $K_m/m$ recovers the Clopper–Pearson interval for the rate.

**Calibration-only intervals (Fig. S6, Table S2).** Data distributions as above; row-simultaneous intervals as described in Section S4; $n \in \{300, 1000\}$ calibration compounds; batch-free correction with $\alpha = \delta = 0.10$; future batch $m = 1000$; 3000 replicates, of which those with fewer than 20 calibration compounds in a class were skipped (103 at $n = 300$, none at $n = 1000$). Class-conditional coverage was evaluated given the realized $m_y$, and mixed-batch coverage with the union over $m_1$. The comparison spends the same budget of $n$ labelled compounds half on calibration and half on an audit set, and computes audit-set intervals for the mixed batch.

**Cost (main text Table S3, Table S3).** Exact population computation, with out-class scores $u \sim \mathrm{Beta}(2, 7)$ and in-class scores $u \sim \mathrm{Beta}(a, b)$. The three settings are prevalence 0.15 with $(9, 3)$ (well separated), prevalence 0.15 with $(4, 3)$ (overlapping), and prevalence 0.5 with $(4, 3)$ (balanced, overlapping). Thresholds $(l, h)$ were searched on a grid of 801 points. Chow: minimum over all $l \le h$. Constrained: minimum subject to class-wise coverage $\ge 0.90$. Default: both coverages exactly 0.90.

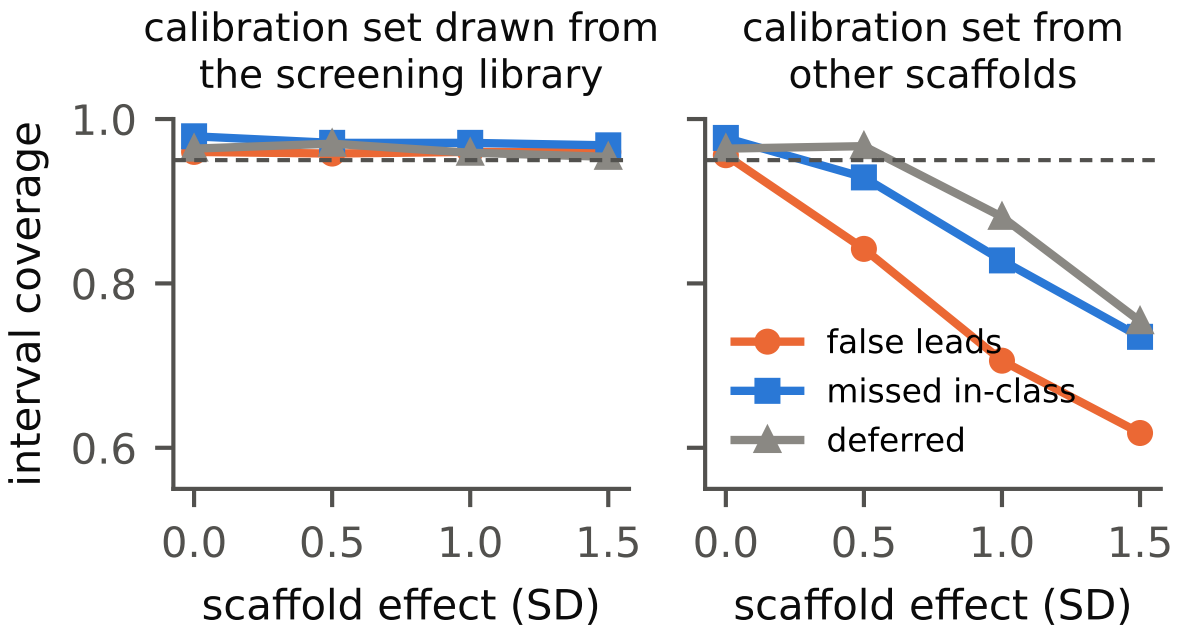



**Figure S4:** Coverage of 95% audit intervals when compounds are clustered in 60 scaffolds that share a shift of the predicted probability (3000 replicates per point). Left: calibration, audit and future compounds drawn from the same scaffolds, as with samples from the screening library. Right: future compounds from new scaffolds.

**Selection.** 4000 replicates. In each, prevalence $\sim$ U(0.1, 0.4), in-class compounds Beta($a$, $b$) with $a \sim$ U(3, 9) and $b \sim$ U(2, 4), out-class compounds Beta(2, 7). There were 500 calibration compounds, 200 test compounds and a future batch of 1000. The candidates were the rules from the standard thresholds at $\alpha_0, \alpha_1 \in \{0.02, \ldots, 0.30\}$ (10 values each), de-duplicated to distinct threshold pairs (median 100, range 90–100). The rule chosen was the one with the fewest audit-set errors (missed in-class compounds plus false leads) among the rules whose one-sided upper limit for deferrals was at most 150. The level of this check was set either at 0.05 for each rule (unadjusted) or at $0.05/J$ with $J$ the number of distinct rules (Bonferroni). For the chosen rule we recorded whether its realized future deferrals exceeded capacity, and whether its two-sided error interval, at the same per-rule level, covered its realized future errors. A rule was found in every replicate. Capacity was exceeded in 0.47% of replicates (unadjusted) and 0.03% (Bonferroni), against the nominal 5%. The chosen rules made 43.5 and 52.5 future errors per batch of 1000. The error interval of the chosen rule covered the realized errors in 95.6% (unadjusted) and 100% (Bonferroni) of replicates. In this design the selection effect on the chosen rule's intervals is small, because neighbouring rules have strongly correlated counts; the unadjusted procedure carries no guarantee nonetheless.

**Tied scores (Table S4).** Out-class scores were drawn from Beta(2, 7) and in-class scores from Beta(6, 3), prevalence 0.2, with 1000 calibration and 1000 future compounds. A fraction $t \in \{0, 4, 6, 10\}\%$ of out-class compounds, in both sets, received one common score at the 0.92 quantile of the out-class distribution. Thresholds were standard at $\alpha = 0.10$, and the class-conditional calibration-only intervals of Section S4 were evaluated with ties kept or broken at random. With ties kept, coverage of the correct-rejection count fell from 0.98 to 0.86, 0.81 and 0.71 for $t = 4$, 6 and 10%, and coverage of the whole out-class row to 0.84, 0.79 and 0.67. With randomized tie-breaking, coverage was at least 0.97 for every cell and at least 0.96 for each row. Tox21 out-class compounds have 3.2–4.4% tied scores (Table S11).

**Scaffold dependence (main text Fig. S4).** There were 60 clusters. Each cluster had a shift $e_1$ of the logit of the prevalence (baseline 0.2) and a shift $e_2$ of the logit of the predicted probability, with $e_1, e_2 \sim \mathrm{N}(0, \sigma^2)$ and $\sigma \in \{0, 0.5, 1, 1.5\}$. Before shifting, scores were drawn from Beta(6, 3) (in-class compounds) and Beta(2, 7) (out-class compounds). There were 1000 calibration and 500 test compounds from one set of clusters. The future batch of 1000 came from the same clusters or from 60 new clusters drawn from the same distribution. Standard thresholds were used at $\alpha = 0.10$, with 3000 replicates per $\sigma$.

**Finite libraries.** To test exchangeability alone, we drew 20 libraries, each of $N = 1000 + n + 1000$ compounds from its own random distribution as above, and held each library fixed. Each library was split 1000 times at random into calibration (1000), test ($n$) and future (1000) sets, which is sampling without replacement from a finite population.

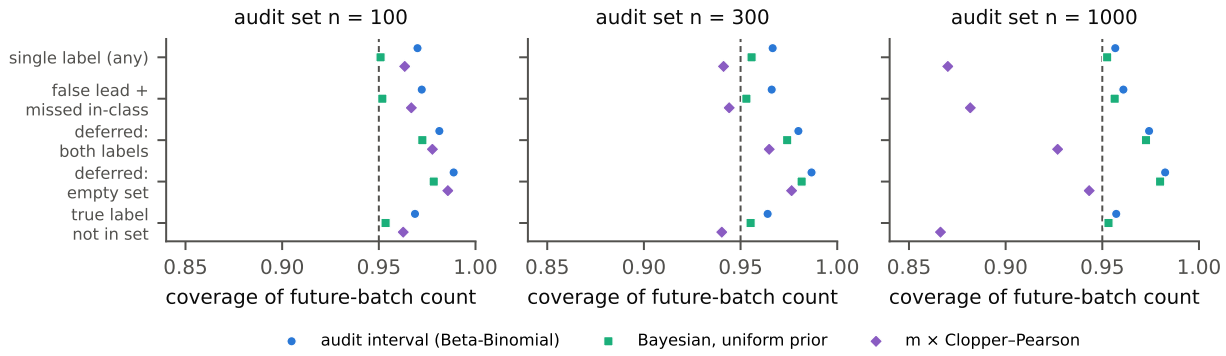



**Figure S5:** Coverage of 95% prediction intervals for the count of each outcome in a future batch of 1000 compounds, from an audit set of $n = 100$, 300 or 1000 compounds (panels); $10^4$ replicates, each drawing its own data distribution. Audit interval: the Beta–Binomial limits of Section 2.5. Bayesian: posterior predictive with a uniform prior. $m\times$Clopper–Pearson: the interval for the rate scaled to the batch, which ignores the sampling variation of the batch. The plug-in Binomial interval, which ignores the uncertainty of the audited rate, covered only 44–93% and is omitted (ESI Table S6). Dashed line: nominal 95%. Monte Carlo standard error is about 0.2%.

**Table S4:** Tied scores and calibration-only intervals ($n = 1000$ calibration and $m = 1000$ future compounds, prevalence 0.2, standard thresholds at $\alpha = 0.10$; 2000 replicates per row). A fraction of out-class compounds share one score at the 0.92 quantile of the out-class distribution, as for a cluster of identical fingerprints. Entries: coverage of the nominal 95% class-conditional intervals, with ties kept (kept) and with randomized tie-breaking (rand.). Row: all cells of the class row covered together.

| Tied | correct rejection | | false lead | | out-class row | | in-class row | |
|---|---|---|---|---|---|---|---|---|
| out-class (%) | kept | rand. | kept | rand. | kept | rand. | kept | rand. |
| 0 | 0.980 | 0.974 | 0.980 | 0.981 | 0.966 | 0.963 | 0.975 | 0.972 |
| 4 | 0.857 | 0.983 | 0.969 | 0.982 | 0.838 | 0.971 | 0.976 | 0.977 |
| 6 | 0.811 | 0.981 | 0.967 | 0.981 | 0.787 | 0.966 | 0.977 | 0.971 |
| 10 | 0.710 | 0.977 | 0.951 | 0.982 | 0.668 | 0.966 | 0.970 | 0.971 |

Such sequences are exchangeable but are not mixtures of independent sequences.[8] Coverage was computed separately for each library (Table S5). The audit interval is valid conditional on each library; the minima of 0.947 are within Monte Carlo error of 0.95. The Bayesian interval fell to 0.935, about two standard errors below nominal.

**Two-stage check of the target.** In a simpler simulation, each of $2 \times 10^5$ replicates drew a rate $p \sim \text{Beta}(a,b)$, a calibration count from $\text{Binomial}(200,p)$, a test count from $\text{Binomial}(n,p)$ and a future count from $\text{Binomial}(m,p)$. Table S7 shows that the audit interval always reaches its nominal level, the Bayesian interval can fall slightly below it, and Clopper–Pearson intervals hold for the long-run rate $p$ but not for its expectation given the calibration data.

# S9 Molecular experiments

**Data and splits, as the main text had them.**

**Simulations.** Eight designs are described in ESI Section S8:

1. audit-set intervals, with $10^4$ replicates, each drawing its own data distribution;
2. finite libraries: 20 fixed libraries, each split at random 1000 times, which gives exchangeable data that are not independent;
3. threshold correction, with absolute Cauchy scores, $n = 50$–$500$, $m = 100$ and $10^6$ draws;
4. expected cost of the swept settings, computed exactly on the population, and selection: 4000 replicates of choosing among 100 settings under an assay capacity, with and without a Bonferroni correction;
5. scaffold dependence: compounds in clusters that share a shift of the predicted probability, with the future batch drawn from the calibration clusters or from new clusters;
6. tied scores: a cluster of identical scores at a threshold, with ties kept or broken at random;

**Table S5:** Finite-library check: coverage of 95% prediction intervals for the future-batch count ($m = 1000$), computed separately for each of 20 fixed libraries (1000 random splits each; Monte Carlo standard error about 0.007 per library). Mean and minimum over libraries.

| $n$ | Outcome | Audit interval | | Bayesian, uniform prior | | Plug-in | |
|---|---|---|---|---|---|---|---|
| | | mean | minimum | mean | minimum | mean | minimum |
| 100 | single label | 0.966 | 0.947 | 0.950 | 0.935 | 0.444 | 0.369 |
| | single-label error | 0.975 | 0.960 | 0.954 | 0.937 | 0.424 | 0.233 |
| | both labels | 0.976 | 0.947 | 0.968 | 0.936 | 0.653 | 0.437 |
| | empty set | 0.989 | 0.960 | 0.982 | 0.943 | 0.791 | 0.416 |
| | miscoverage | 0.969 | 0.955 | 0.953 | 0.936 | 0.443 | 0.425 |
| 300 | single label | 0.966 | 0.947 | 0.955 | 0.943 | 0.660 | 0.625 |
| | single-label error | 0.967 | 0.951 | 0.954 | 0.939 | 0.657 | 0.468 |
| | both labels | 0.980 | 0.947 | 0.975 | 0.943 | 0.810 | 0.653 |
| | empty set | 0.986 | 0.957 | 0.981 | 0.944 | 0.850 | 0.625 |
| | miscoverage | 0.961 | 0.951 | 0.952 | 0.939 | 0.660 | 0.622 |

**Table S6:** Coverage of 95% prediction intervals for the count of each outcome in a future batch of $m = 1000$, and mean interval width, from an audit set of $n$ compounds; 10000 replicates per audit size. Audit interval: the Beta–Binomial limits of Section S3. $m\times$CP: $m$ times the Clopper–Pearson interval for the rate, rounded outward. Plug-in: Binomial$(m, K/n)$ quantiles. Last column: Clopper–Pearson coverage of the long-run rate $p_g(Q)$, the quantity it is designed for.

| $n$ | Outcome | Audit interval | | Bayesian, uniform prior | | $m\times$CP | | Plug-in | CP, rate |
|---|---|---|---|---|---|---|---|---|---|
| | | coverage | width | coverage | width | coverage | width | coverage | coverage |
| 100 | single label | 0.970 | 128.0 | 0.951 | 121.0 | 0.963 | 123.5 | 0.436 | 0.970 |
| | single-label error | 0.972 | 102.0 | 0.952 | 96.1 | 0.967 | 98.7 | 0.437 | 0.973 |
| | both labels | 0.981 | 102.0 | 0.973 | 97.0 | 0.978 | 98.4 | 0.689 | 0.981 |
| | empty set | 0.989 | 64.0 | 0.978 | 61.0 | 0.986 | 62.1 | 0.747 | 0.989 |
| | miscoverage | 0.969 | 121.4 | 0.954 | 114.2 | 0.963 | 117.2 | 0.448 | 0.969 |
| 300 | single label | 0.967 | 78.2 | 0.956 | 75.5 | 0.941 | 70.1 | 0.661 | 0.965 |
| | single-label error | 0.966 | 61.2 | 0.953 | 58.8 | 0.944 | 55.3 | 0.662 | 0.966 |
| | both labels | 0.980 | 58.2 | 0.974 | 56.6 | 0.965 | 52.6 | 0.812 | 0.980 |
| | empty set | 0.987 | 32.9 | 0.982 | 31.9 | 0.976 | 30.5 | 0.849 | 0.985 |
| | miscoverage | 0.964 | 74.2 | 0.955 | 71.3 | 0.940 | 66.7 | 0.663 | 0.962 |
| 1000 | single label | 0.957 | 51.7 | 0.953 | 50.9 | 0.870 | 37.9 | 0.845 | 0.957 |
| | single-label error | 0.961 | 40.4 | 0.956 | 39.6 | 0.882 | 29.9 | 0.849 | 0.958 |
| | both labels | 0.974 | 36.8 | 0.973 | 36.4 | 0.927 | 27.0 | 0.915 | 0.976 |
| | empty set | 0.983 | 19.9 | 0.980 | 19.5 | 0.943 | 14.9 | 0.930 | 0.981 |
| | miscoverage | 0.957 | 49.3 | 0.953 | 48.4 | 0.866 | 36.2 | 0.845 | 0.954 |

7. calibration-only intervals: 3000 replicates per calibration size, compared with spending the same labelling budget half on calibration and half on an audit set.

8. a two-level draw, in which a rate is drawn and then the calibration, audit and future counts are drawn given it, used to separate a statement about the long-run rate from a statement about the realized count of one batch.

**Data.** Molecules were standardized with RDKit[18], which keeps the largest fragment, neutralizes it and writes canonical SMILES; duplicates were then merged. One model was used throughout: Morgan fingerprints[19] (radius 2, 2048 bits) and CatBoost gradient-boosted trees[20] (500 trees, depth 6, learning rate 0.1) with fixed hyperparameters. The experiments concern the coverage settings and their audited consequences for a given model, not the choice of model. For the twelve Tox21 endpoints,[21,22] each endpoint was split 50/25/25 into training, calibration and audit sets: 50 times at random and 20 times by Bemis–Murcko scaffold.[23] Under the scaffold splits we also drew pilot samples of 200, 500 and 1000 compounds at random from the audit scaffolds, used each as the calibration set, and screened the remaining compounds. For AqSolDB,[24] compounds with $\log S \geq -4$ were labelled soluble. Ten stratified 70/15/15 splits were made, the model was trained on all training compounds, and calibration and audit were restricted to $\log P > 3.5$. Solubility most often limits lipophilic compounds, so a screen of this subpopulation is a realistic use; it is also a subpopulation on which the model's scores are distributed differently from the training data. For CALiSol-23,[25] a compilation of measured conductivities of lithium-salt electrolytes, each formulation (salt, concentration and solvent fractions) was labelled good if its conductivity at 20–30 °C was at least 8 mS cm$^{-1}$. The same model was trained on all temperatures of the training formulations, with temperature, solvent fractions and fraction-weighted solvent descriptors as inputs, and scored each formulation at 25 °C. Four designs, 50 repeats each, drew calibration and audit sets from random

**Table S7:** Two-stage simulation ($2 \times 10^5$ replicates, 95% nominal).

| $n$ | $m$ | $p \sim$ | Audit interval | Bayes (uniform) | Bayes (Jeffreys) | CP for $p$ | CP for $\mathrm{E}[p \mid \mathscr{D}_{\mathrm{cal}}]$ |
|---|---|---|---|---|---|---|---|
| 46 | 100 | Beta(1,30) | 0.993 | 0.985 | 0.979 | 0.986 | 0.975 |
| 46 | 1000 | Beta(1,30) | 0.987 | 0.948 | 0.965 | 0.986 | 0.974 |
| 100 | 1000 | Beta(1,60) | 0.986 | 0.949 | 0.971 | 0.986 | 0.971 |
| 235 | 1000 | Beta(2,40) | 0.970 | 0.955 | 0.953 | 0.968 | 0.880 |
| 500 | 1000 | Beta(2,5) | 0.958 | 0.953 | 0.953 | 0.957 | 0.727 |

formulations, from held-out salt–solvent systems, or, for a new salt, from the other salts or from a pilot sample of the new salt. To check the audit intervals on real data, each audit set was split at random, 20 times, into an audit half and a stand-in future batch. A random halving makes the two halves exchangeable by construction, so these checks verify that the construction and its implementation are right and cannot fail for any model, representation or label noise; they are not evidence that an audit prices a batch assembled some other way. The designs that test that are the block holdouts and the declared round, where the batch is a distinct set of molecules. For the DNA-encoded library,[26] a sample of one million molecules was taken from the released table and a molecule was labelled in-class if it carried the library's own binding call for soluble epoxide hydrolase, a rate of 0.74%; the analysis runs on a further random subsample of 200 000 of those molecules, at a rate of 0.72%. Each molecule is built from one building block at each of three positions, so designs were defined over blocks rather than molecules: assignment at random, and assignment of whole blocks at one of the three positions, so that a batch molecule carries a block the training molecules do not. A declared round, 30% of it built on first-position blocks absent from training, was then prepared for in three ways that differ only in where the calibration and audit sets draw their novel chemistry. Counts for this library are for the realized batch of each repeat, tens of thousands of candidates, because at a sub-percent rate a batch of 1000 carries too few in-class molecules for a confirming setting to be bounded away from zero; the scripts additionally scale each rate to a nominal $10^5$ so repeats can be compared, and those scaled figures are not the ones quoted here. The thresholds for this library were planned at $m = 1000$ rather than at the realized batch size, which makes the deployed rule more conservative than the request rather than less; ESI Section S14 gives the consequence, and it should be repaired before the library results are relied on. A further division of the electrolyte formulations by the paper each measurement came from, rather than at random, is described in ESI Section S10. Details, counts and seeds are in ESI Section S9, and for the library in ESI Section S14.

**Label problems, by dataset.** Tox21 labels are missing for 7.6–26.0% of the 7592 compounds depending on the endpoint, and the missingness is not random: the 2926 compounds with no missing endpoint are active in 2.7% of their labelled cells, the 3188 missing one to three in 8.9%, and the 1478 missing four or more in 23.2%. The blanks are the inconclusive calls of the original screen, which fall disproportionately on cytotoxic and interfering compounds — the kind a real primary screen contains and a model scores high. Every Tox21 audit set here is therefore a random sample of the *labelled* subpopulation rather than of the library, at a lower in-class rate than the library carries. That is an instance of this paper's own thesis rather than an exception to it, and it means the Tox21 results should be read as verification of the construction on a clean pool rather than as evidence about a screen of the whole collection. AqSolDB combines measurements from many sources, with a typical spread of 0.5–0.7 log units between sources;[24] labels near the cut-off at $\log S = -4$ are therefore noisy, and at that spread the expected label-flip rate at the cut is 8–9% among the lipophilic compounds screened, which is the size of the coverage requests being swept; Section 3.4 measures what that costs, and the scenario is the part it costs the most; the curation rule also removed the most discordant duplicates, which a real batch would contain. CALiSol-23 likewise pools 27 sources, and several formulations were measured at many temperatures or concentrations, so neighbouring formulations are correlated. Its label also carries the provenance of the measurement rather than only the chemistry: 90% of room-temperature formulations have a single measurement in the 293–303 K window, and the temperature of that measurement is bimodal, 365 below 296 K and 917 above 300 K against 200 near 298 K, while conductivity rises by a median $0.136\,\mathrm{mS\,cm^{-1}\,K^{-1}}$, so the window spans 17% of the $8\,\mathrm{mS\,cm^{-1}}$ cut and a quarter of the formulations sit within 20% of it. Label temperature is moreover confounded with salt and with source, so the salt holdouts change the measurand as well as the chemistry. Two entries of the salt column, LiTFSI and $\mathrm{LiN(CF_3SO_2)_2}$, are the same compound under two names, 27 and 9 room-temperature rows whose mean label temperatures differ by five degrees; they were encoded separately and can therefore be split across a salt holdout. Both should be repaired before the electrolyte results are relied on: the first by interpolating each formulation to a common temperature, the second by merging the names. The guarantees refer to the labels as recorded.

**Data.** Tox21 was taken from the MoleculeNet file distributed with DeepChem, and AqSolDB from the curated set released by its authors;[24] checksums are listed with the scripts. Molecules were parsed with RDKit,[18] reduced to the largest fragment, neutralized where possible and written as canonical SMILES; molecules that failed or had fewer than two heavy atoms were removed. For Tox21, 8014 records gave 7592 unique compounds. Duplicate labels were merged

per endpoint, and 139 conflicting endpoint labels were set to missing. For AqSolDB, 9982 records gave 9399 unique compounds. Duplicates were averaged when their $\log S$ values differed by at most 0.5 log units and removed otherwise, leaving 9254 compounds, 2467 of them with RDKit $\log P > 3.5$.

**Features and models.** Morgan fingerprints[19] had radius 2 and 2048 bits. CatBoost[20] used 500 trees, depth 6, learning rate 0.1 and default regularization; hyperparameters were fixed in advance and not tuned. There was no early stopping, and hence no tuning set. A single model class and representation were used throughout, because the comparisons in this paper are between ways of obtaining intervals for a given model, not between models.

**Tox21 splits.** Each endpoint was split into training, calibration and audit sets in proportions 50/25/25, 50 times at random. It was also split 20 times by Bemis–Murcko scaffold.[23] For the scaffold splits, scaffolds were shuffled and each scaffold group was assigned to the set furthest below its target size; acyclic compounds were treated as singleton groups. Training and audit sets shared no scaffold. One model was fitted per split. Scores of calibration and test compounds were tie-broken at random before thresholds were set (Section S2); a median of 3.3–4.3% of scores per endpoint were tied, and in random splits a median of 3.5% of test compounds (at most 4.8%) had a fingerprint identical to one in the calibration set. Such duplicates do not break exchangeability under random splits, because which member of a pair falls on which side is itself random. Thresholds used $\alpha = \delta = 0.10$ for both classes. The batch correction took $m_y$ from the calibration prevalence and the audit-set size, not from the realized test class counts. Standard, batch-free and one-sided DKW thresholds were computed on the same splits. Pilot samples of 200, 500 and 1000 compounds were drawn at random from the audit set of each scaffold split (1000 only when at least 1200 test compounds were available, giving 224 pilots); each pilot was used as the calibration set and the remaining test compounds as the screened batch. Pilots with fewer than five in-class compounds were skipped (23 of 240 of 200 compounds, 1 of 240 of 500, none of 1000). Seeds: $100000 + 1000j + r$ (random) and $100000 + 1000j + r + 500$ (scaffold), for endpoint $j$ and repeat $r$. These runs replace those of the first version of this paper, which used other seeds and no tie-breaking; scores, labels and split indices of every run are stored.

**Violations on real data.** For each split and class, the exact probability that per-class test coverage falls below 90% follows from the batch law for the thresholds used and the realized test class count (Section S2). Summing these probabilities gives the expected number of violations, and the exact distribution of the number of violations is Poisson-binomial. Over the 600 random splits, the batch correction gave 55 violations for out-class compounds (56.2 expected; one-sided $p = 0.58$) and 45 for in-class compounds (42.3; $p = 0.35$). Per endpoint (Table S12), the largest excess was SR-MMP out-class compounds, 9 of 50 against 4.6 ($p = 0.039$ unadjusted; not significant after correction for 24 tests). In the first version of this paper, with other seeds and without tie-breaking, the out-class count was 74 of 600 against 56.6 expected ($p = 0.011$). The present run, with fresh seeds, does not reproduce that excess, so it was most probably a chance finding; we know of no mechanism by which refitting the model could create an excess, because each split's violation probability given its fitted model is exactly the theoretical value. Calibrating on other scaffolds gave 67 and 44 violations in 240 scaffold splits, against 22.8 and 18.2 expected ($p < 10^{-6}$).

**Widths on real data.** For each random split, calibration-only intervals for a batch of 1000 were compared with audit-set intervals computed from the audit set of the same split, which is as large as the calibration set. Median width ratios per endpoint were 1.42–1.70 for correct rejections, 1.42–1.50 for false leads, 1.62–1.81 for confident hits, 1.50–1.82 for missed in-class compounds, 2.27–2.44 for deferred out-class compounds and 1.93–2.46 for deferred in-class compounds. The comparison is not budget-matched: it sets the calibration-only intervals against an audit set that would need as many extra labels as calibration.

**Pilot samples (Table S8).** A request is infeasible when even the most conservative threshold ($u = 1$) does not meet it, which at $\alpha = \delta = 0.10$ happens with fewer than 22–23 calibration compounds of the class, for any batch size from 20 to 5000 (22 in the batch-free limit, since $1 - 0.9^{22} \geq 0.9$). The software then uses the most conservative threshold, and the request is not certified. Violations among feasible pilots followed the exact law except for out-class compounds with pilots of 200 (11 of 65 against 5.1, $p = 0.013$, one of six tests).

**AqSolDB splits.** Ten stratified 70/15/15 splits were made, with seeds $7000 + r$. The model was trained on all training compounds. Calibration and screening used compounds with $\log P > 3.5$: about 370 per set, 22% soluble. For the coverage request sweep, $\alpha_0, \alpha_1 \in \{0.025, 0.05, \ldots, 0.30\}$ and $\delta = 0.10$. The batch correction used $m_y$ equal to the expected number of class-$y$ compounds in a batch of 1000 at the calibration prevalence. Requests that selected the same pair of order statistics were merged. Scores were tie-broken at random (seed $7000 + r$) before any threshold was set. The Pareto front, the cost map and the regions of Fig. 3 and Table 4 are for split 0; the same quantities for every split are in Table S10. The robust region uses Clopper–Pearson intervals for the within-set fractions from the audit

**Table S8:** Tox21 scaffold splits: calibration on scaffolds other than those screened, or on a random pilot sample of the screened compounds (pilot sizes 200, 500, 1000; the rest of the test set is screened). Pilots with fewer than five in-class were skipped. Infeasible: pilots for which the in-class-class request ($\alpha = \delta = 0.10$ for the planned number of in-class) cannot be met, so the most conservative threshold is used. Violations: splits in which test coverage of the class fell below 90%, observed/expected under the exact law; for feasible pilots only, with their number. Last columns: coverage (%) of calibration-only intervals, class-conditional and for the mixed screened batch.

| Calibration | pilots | skipped | median in-class | infeasible (%) | violations out-class | violations in-class | in-class, feasible pilots only | calibration-only class | calibration-only mixed |
|---|---|---|---|---|---|---|---|---|---|
| other scaffolds | – | – | – | – | 67/22.8 | 44/18.2 | – | 74.2 | 80.1 |
| 200 | 240 | 23 | 13 | 70 | 20/17.4 | 62/55.9 | 3/3.8 (65) | 99.0 | 99.8 |
| 500 | 240 | 1 | 29 | 41 | 15/20.9 | 31/28.5 | 13/9.7 (141) | 99.0 | 100.0 |
| 1000 | 224 | 0 | 54 | 11 | 23/21.1 | 27/19.6 | 18/16.1 (199) | 99.3 | 100.0 |

**Table S9:** Monotonicity diagnostic for the Tox21 models: percentage of in-class in ten equal-count bins of the test-set score $u$, pooled over random splits, from the lowest to the highest bin; number of adjacent bins in which the fraction decreases; largest such decrease in standard errors ($z$). With ten bins and noisy low-prevalence bins, decreases with $z < 2$ are expected by chance.

| Endpoint | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | decr. | max $z$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NR-AR | 2.7 | 2.6 | 2.2 | 1.7 | 1.8 | 1.7 | 1.5 | 1.9 | 2.2 | 23.1 | 5 | 2.4 |
| NR-AR-LBD | 1.5 | 1.3 | 1.2 | 1.2 | 1.2 | 1.1 | 1.2 | 1.0 | 1.4 | 24.9 | 5 | 1.3 |
| NR-AhR | 1.2 | 0.9 | 0.9 | 1.8 | 3.9 | 5.6 | 8.2 | 13.0 | 24.0 | 58.4 | 2 | 1.8 |
| NR-Aromatase | 1.8 | 1.9 | 2.1 | 2.6 | 3.2 | 3.5 | 4.0 | 4.8 | 6.1 | 19.8 | 0 | 0.0 |
| NR-ER | 7.4 | 7.0 | 6.7 | 6.9 | 7.5 | 8.1 | 10.2 | 11.5 | 16.6 | 43.7 | 2 | 0.9 |
| NR-ER-LBD | 2.4 | 2.0 | 2.5 | 1.9 | 2.4 | 2.5 | 2.7 | 3.1 | 4.1 | 27.5 | 2 | 2.6 |
| NR-PPAR-$\gamma$ | 1.7 | 1.5 | 1.3 | 1.5 | 1.6 | 1.8 | 1.7 | 2.2 | 2.8 | 12.6 | 3 | 1.0 |
| SR-ARE | 4.0 | 4.0 | 5.1 | 7.5 | 9.2 | 11.4 | 15.2 | 20.6 | 30.3 | 51.4 | 1 | 0.1 |
| SR-ATAD5 | 1.1 | 1.5 | 1.3 | 1.5 | 2.3 | 2.3 | 3.1 | 4.0 | 4.8 | 16.4 | 1 | 0.8 |
| SR-HSE | 2.9 | 3.3 | 3.4 | 3.1 | 3.2 | 3.6 | 4.5 | 5.1 | 7.3 | 20.8 | 1 | 1.1 |
| SR-MMP | 1.7 | 2.6 | 2.9 | 4.1 | 4.8 | 7.0 | 11.0 | 18.5 | 35.1 | 70.9 | 0 | 0.0 |
| SR-p53 | 1.7 | 1.8 | 2.1 | 2.8 | 3.5 | 4.1 | 4.9 | 7.0 | 10.4 | 25.8 | 0 | 0.0 |

set, Bonferroni over the sets that occur (Section S7). In the main text, the Pareto front, settings A and B, the cost map and the regions use the audited (audit-set) counts of each setting. The calibration-only capacity rule in Table S10 uses calibration-only upper limits for the summed deferral cells at level $0.05/144$ for each request. Table S13 compares the cost of the setting chosen from calibration estimates with Chow's rule on recalibrated probabilities, and is discussed in the main text: the swept setting is indistinguishable from the isotonic version and dearer than the Platt version, which is why the case made for the sweep is that it needs no cost ratio rather than that it is cheaper when one is available. Cost baselines were fitted on the full calibration set: Platt scaling as a logistic regression on the logit of $u$; isotonic regression; and the two thresholds on $u$, from a grid of 201 calibration quantiles, with the lowest calibration cost. The set rule was chosen among all swept rules and three trivial rules (defer all, label all out-class, label all soluble) by its cost estimated from the calibration set; the trivial rule was chosen in 4 of 10 splits at $\lambda = 50$, $\rho = 0.5$ and never otherwise. Restricting the choice to rules from requests with $\alpha_0, \alpha_1 \leq 0.10$ changed the mean cost by at most 0.011 per compound.

**Audit check on real data (main text Table 3).** For every Tox21 run (600 random, 240 scaffold), scores were tie-broken, the batch-correction thresholds were set on the calibration set, and the audit set was split 20 times at random into two halves of equal size. Intervals for each outcome's count in one half (the stand-in future batch) were computed from the other half at $\gamma = 0.05$. For AqSolDB, the same was done for every distinct setting of the sweep in each of ten splits (20 halvings each, 216,960 checks), pointwise and with Bonferroni over the 144 requests. In each halving, a capacity rule was also chosen on the audit half: the setting with fewest audited errors (missed solubles plus false leads) whose one-sided upper limit for deferrals, at level $0.05/144$, was at most half the stand-in batch. Its realized deferrals in the stand-in batch never exceeded capacity in 200 halvings; the median upper limit was 484 and the median realized count 304 per 1000. Per endpoint, coverage of the Tox21 audit intervals was 95.8–97.1% (random) and 95.0–97.5% (scaffold), averaged over outcomes.

**CALiSol-23 (main text Section 3.6).** The file (13,825 rows; checksum with the scripts) lists, per measurement, the source DOI, conductivity $\kappa$ (mS cm$^{-1}$), temperature, salt, concentration with its unit (mol/kg or mol/L), the basis of the solvent ratios (weight, volume or mole) and the fractions of 38 solvents.[25] Rows without $\kappa$ (523) were removed,

**Table S10:** AqSolDB, per split. Rules: distinct rules from the sweep; Pareto: rules not dominated on estimated missed solubles, false leads and deferrals; cheapest: rules that are cheapest for some cost ratio on the grid of Fig. 3; regions: percentage of that grid where acting on single labels is cheapest for the cheapest rule, at the estimated within-set fractions (point) and for all fractions in simultaneous 95% test-set intervals (robust). Pareto, cheapest and regions use the audit-set rates of each rule. Capacity (calibration only): at most 500 deferrals per 1000; the rule with fewest estimated errors whose calibration-only upper limit for deferrals is within capacity, Bonferroni over the 144 requests; entries are upper limit / realized audit-set deferrals / realized audit-set errors, per 1000.

| Split | rules | Pareto | cheapest | region (%) | | capacity rule |
|---|---|---|---|---|---|---|
| | | | | point | robust | (calibration only) |
| 0 | 144 | 104 | 25 | 24.7 | 2.1 | 491 / 174 / 143 |
| 1 | 132 | 96 | 48 | 57.1 | 1.9 | 488 / 174 / 123 |
| 2 | 144 | 74 | 27 | 53.1 | 2.5 | 489 / 146 / 104 |
| 3 | 132 | 92 | 26 | 32.9 | 2.5 | 500 / 233 / 84 |
| 4 | 132 | 84 | 23 | 48.6 | 4.9 | 490 / 128 / 161 |
| 5 | 132 | 107 | 36 | 39.0 | 5.2 | 492 / 189 / 90 |
| 6 | 144 | 120 | 26 | 33.8 | 6.1 | 490 / 125 / 151 |
| 7 | 132 | 96 | 28 | 64.7 | 3.8 | 493 / 177 / 156 |
| 8 | 132 | 109 | 34 | 46.6 | 0.6 | 495 / 101 / 151 |
| 9 | 132 | 108 | 39 | 21.9 | 0.9 | 498 / 153 / 108 |

and so were the 23 rows of DOI 10.1007/s10008-002-0300-9, whose values for 1 mol/kg $LiPF_6$ in carbonate mixtures lie about 100 times below those of every other source. Salt names were stripped of whitespace and solvent fractions normalized to sum to one. A formulation is the salt, the concentration and its unit, the ratio basis and the solvent fractions rounded to three decimals (5143 formulations, 139 salt–solvent systems, 26 sources); the precision matters, since rounding to four decimals instead splits the pool into 1522 room-temperature formulations rather than 1495. The screened quantity is the median $\kappa$ of a formulation over its measurements at 293.15–303.15 K (1495 formulations); the label is 1 if it is at least 8 mS cm$^{-1}$ (275 formulations). Features: salt one-hot, concentration, unit and basis flags, the 38 solvent fractions, fraction-weighted RDKit descriptors of the solvents (molecular weight, log P, polar surface area, hydrogen-bond acceptors, fraction of $sp^3$ carbon, rotatable bonds, molar refractivity; solvents mapped to SMILES in the scripts), and temperature. CatBoost (as for the other datasets) was trained on every row, at any temperature, of the training formulations, and scored the room-temperature formulations at 298.15 K. Designs (seeds $30000 + 1000d + r$): *random*, room-temperature formulations split 50/25/25 into training, calibration and audit; *system*, whole salt–solvent systems assigned to the three sets, with no row of a held system in training (one of 50 splits skipped for having fewer than five good calibration formulations); *new salt*, the model trained on two thirds and calibrated on one third of the formulations of all salts other than $LiPF_6$, and audited on all 606 $LiPF_6$ formulations (154 good); *pilot*, the same model calibrated on a random 40% of the $LiPF_6$ formulations and audited on the rest. Scores were tie-broken at random. Thresholds used the batch correction at $\alpha = \delta = 0.10$ with $m_y$ planned from the calibration prevalence and the audit-set size. Audit checks, the sweep ($12 \times 12$ requests) and the capacity rule (at most half the stand-in batch, chosen on one half with Bonferroni over 144 requests) followed the protocol of the audit check above, with 20 halvings per split. For the new salt, a median of 41 good formulations calibrated the good class, all requests were attainable, and coverage of the good class averaged 29%; with the pilot, 62.5 good formulations calibrated it and coverage averaged 95%. The poor-class excess with the pilot (8 against 4.0 expected) has one-sided Poisson-binomial $p = 0.04$.

**Real-score check of the batch law.** For each endpoint, one CatBoost model was trained on a random half of the labelled compounds. The other half was re-split 5000 times into calibration and test halves, with class counts held fixed; the batch law is conditional on class counts. Thresholds came from the batch correction ($\alpha = \delta = 0.10$, $m_y$ = test compounds of class $y$). Ties were kept in this check; for the one-sided coverage event they can only make violations rarer. Table S11 compares the observed fraction of re-splits in which per-class test coverage fell below 90% with the exact Beta–Binomial value. Pooled over endpoints, the observed rates were 9.45% against 9.35% for out-class compounds ($z = 0.79$, one-sided $p = 0.22$) and 7.92% against 7.89% for in-class compounds ($z = 0.30$, $p = 0.38$). This check verifies the implementation and the law on real scores; it is not an explanation of chance variation in other runs. The largest per-endpoint deviations were NR-PPAR-$\gamma$ out-class compounds (10.58% against 9.63%), SR-ATAD5 out-class compounds (10.30% against 9.52%) and NR-ER in-class compounds (8.04% against 9.80%), among 24 comparisons. The first two are about two Monte Carlo standard errors (0.42 percentage points) above the exact value; the third is about four standard errors below it, in the conservative direction, as ties can make it.

**Table S11:** Batch coverage law on real Tox21 scores: percentage of 5000 re-splits in which per-class test coverage fell below 90%, observed and exact, with calibration size $n$ and the percentage of tied out-class scores.

| | Out-class | | | | In-class | | |
|---|---|---|---|---|---|---|---|
| Endpoint | $n$ | observed | exact | ties | $n$ | observed | exact |
| NR-AR | 1693 | 10.12 | 9.71 | 3.4 | 61 | 8.66 | 8.56 |
| NR-AR-LBD | 1583 | 8.68 | 9.02 | 4.4 | 53 | 5.22 | 5.63 |
| NR-AhR | 1394 | 10.08 | 9.70 | 3.5 | 187 | 9.52 | 9.35 |
| NR-Aromatase | 1332 | 9.26 | 9.17 | 3.9 | 72 | 5.38 | 5.09 |
| NR-ER | 1298 | 9.34 | 9.89 | 3.4 | 196 | 8.04 | 9.80 |
| NR-ER-LBD | 1597 | 8.98 | 9.00 | 4.1 | 88 | 6.62 | 6.60 |
| NR-PPAR-$\gamma$ | 1524 | 10.58 | 9.63 | 3.5 | 39 | 6.08 | 5.77 |
| SR-ARE | 1186 | 8.88 | 8.94 | 3.8 | 225 | 9.26 | 8.68 |
| SR-ATAD5 | 1649 | 10.30 | 9.52 | 4.1 | 63 | 8.00 | 8.19 |
| SR-HSE | 1475 | 8.90 | 9.14 | 3.2 | 93 | 8.10 | 8.14 |
| SR-MMP | 1169 | 8.90 | 8.84 | 3.3 | 237 | 9.90 | 9.26 |
| SR-p53 | 1541 | 9.36 | 9.68 | 4.1 | 100 | 10.24 | 9.56 |

**Table S12:** Tox21 endpoints and results over 50 random splits with the batch correction ($\alpha = \delta = 0.10$ per class) unless stated. Columns: labelled compounds; in-class (%); median calibration in-class; median test AUROC (random, scaffold splits); median single-label fraction (%) with the batch correction and with the standard threshold; fraction of in-class among single-label in-class calls (%) and enrichment over prevalence; per-class violations, i.e. splits in which test coverage of the class fell below 90%, observed/expected under the exact law (out-class, in-class); coverage of the class-conditional calibration-only intervals (%); median tied scores before tie-breaking (%).

| Endpoint | $n$ | act. | cal. act. | AUC r | AUC s | single | single (std) | hit | enr. | viol. 0 | viol. 1 | cal.-only | ties |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NR-AR | 7017 | 4.1 | 74 | 0.76 | 0.71 | 18.3 | 30.7 | 22 | 5.3 | 6/4.9 | 6/3.1 | 97.0 | 4.0 |
| NR-AR-LBD | 6545 | 3.6 | 57 | 0.83 | 0.80 | 16.0 | 33.0 | 23 | 6.5 | 1/4.5 | 2/3.0 | 99.7 | 4.3 |
| NR-AhR | 6328 | 11.8 | 186 | 0.87 | 0.85 | 61.9 | 71.8 | 48 | 4.1 | 6/4.8 | 3/4.3 | 98.7 | 3.9 |
| NR-Aromatase | 5619 | 5.0 | 72 | 0.73 | 0.72 | 20.6 | 35.8 | 20 | 4.0 | 5/4.5 | 2/3.1 | 99.3 | 4.0 |
| NR-ER | 5976 | 12.6 | 188 | 0.70 | 0.69 | 22.6 | 30.6 | 39 | 3.1 | 7/4.8 | 6/4.3 | 99.3 | 3.5 |
| NR-ER-LBD | 6739 | 5.1 | 85 | 0.78 | 0.74 | 21.6 | 33.0 | 26 | 5.0 | 3/4.5 | 4/3.2 | 99.0 | 4.0 |
| NR-PPAR-$\gamma$ | 6253 | 2.9 | 44 | 0.72 | 0.72 | 13.6 | 28.1 | 13 | 4.6 | 3/4.8 | 2/2.0 | 99.0 | 4.1 |
| SR-ARE | 5647 | 15.9 | 228 | 0.77 | 0.73 | 37.0 | 48.4 | 47 | 3.0 | 4/4.6 | 4/4.2 | 99.3 | 3.3 |
| SR-ATAD5 | 6850 | 3.8 | 66 | 0.76 | 0.68 | 23.4 | 38.1 | 16 | 4.3 | 4/4.7 | 5/3.3 | 99.0 | 4.0 |
| SR-HSE | 6272 | 5.7 | 88 | 0.70 | 0.67 | 20.2 | 28.6 | 21 | 3.6 | 1/4.6 | 4/3.6 | 98.3 | 3.3 |
| SR-MMP | 5627 | 15.9 | 222 | 0.86 | 0.82 | 54.3 | 67.3 | 58 | 3.6 | 9/4.6 | 3/4.2 | 98.0 | 3.7 |
| SR-p53 | 6564 | 6.4 | 102 | 0.77 | 0.72 | 28.8 | 43.0 | 25 | 3.9 | 6/4.8 | 4/4.1 | 98.0 | 3.7 |

# S10 Dividing an audit from a batch by provenance

The electrolyte compilation records a DOI for every measurement, which makes a division by source possible. Formulations were reduced to one row each as elsewhere, the room-temperature window and the $8\,\mathrm{mS\,cm^{-1}}$ cut were applied unchanged, and 40% of the formulations were assigned to training, 20% to calibration and the remaining 40% to the divisions below. The model was fitted on every temperature row of the training formulations, with the standard representation and hyperparameters of this work (500 trees, depth 6, learning rate 0.1), is therefore disjoint from the calibration set that sets its thresholds, and was frozen before any division of the remainder. Ten repeats were run under seeds $777 + r$, with the tie-breaking jitter under $31337 + r$. An earlier version of this experiment fitted the model on its own calibration set, which voids the coverage guarantee for the design and makes it incomparable with the rest of the work; every figure reported here is from the corrected run, in which the losses are real but smaller.

The remainder was then divided three ways, the same model and calibration set serving all three. *Halved* splits it at random into an audit half and a stand-in batch, which is the check used throughout the main text. *By source* partitions the papers at random into two groups and lets each formulation follow its own paper, so that no paper contributes to both sides. *One source out* takes a single paper's formulations as the batch and all others as the audit, repeated over the five papers contributing at least 40 formulations to the remainder, giving 41 runs over the ten repeats; because the inference is about papers, coverage is summarized with the paper as the unit, over the five per-paper means.

Coverage of the six consequence lines, with the repeat as the unit, was $0.97 \pm 0.04$ halved, $0.70 \pm 0.12$ by source, and $0.77 \pm 0.11$ one source out with the paper as the unit over five per-paper means running from 0.58 to 0.90. Median audit-to-batch ratios of the in-class rate were 1.02, 0.67 and 1.88, with per-repeat ratios spanning 0.32 to 2.24 and 0.36 to 8.14 in the two source divisions, so the division moves prevalence as well as the mapping from score to outcome

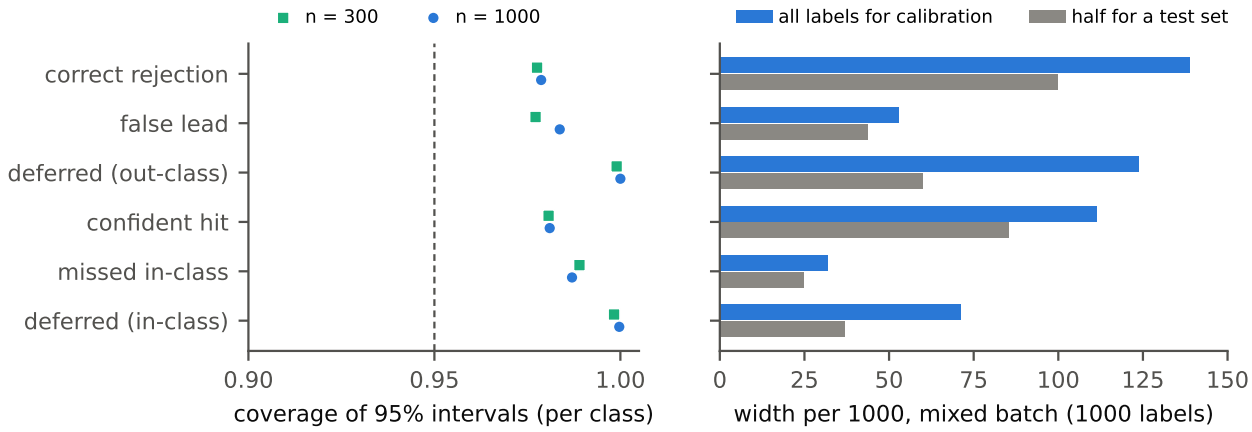



**Figure S6:** Intervals from the calibration set alone. Left: coverage of 95% intervals for each cell's count among the future compounds of that class (3000 replicates; dashed line 95%). Right: mean interval width for a mixed batch of 1000 compounds, when a budget of 1000 labelled compounds is used entirely for calibration or split equally between calibration and an audit set.

**Table S13:** AqSolDB lipophilic screen: mean test-set cost per compound (units of a wasted follow-up) over 10 splits. All baselines were fitted on the full calibration set and applied to the test set. Chow: cost-optimal action per compound on probabilities recalibrated by Platt scaling (logistic regression on the logit of $u$) or isotonic regression. Two thresholds: the two thresholds on $u$ minimizing empirical calibration cost. Set rule: cheapest candidate on calibration estimates, among the swept rules and the trivial rules (defer all, label all out-class, label all soluble). Last two columns: paired difference, set rule minus Chow, mean (standard error over splits).

| $\lambda$ | $\rho$ | Chow (Platt) | Chow (isotonic) | two thresholds | set rule | defer all | set − Platt | set − isotonic |
|---|---|---|---|---|---|---|---|---|
| 5 | 0.3 | 0.247 | 0.252 | 0.252 | 0.258 | 0.300 | +0.011 (0.003) | +0.006 (0.004) |
| 10 | 0.5 | 0.414 | 0.419 | 0.420 | 0.432 | 0.500 | +0.018 (0.010) | +0.013 (0.009) |
| 20 | 0.8 | 0.622 | 0.660 | 0.661 | 0.663 | 0.800 | +0.041 (0.019) | +0.002 (0.023) |
| 50 | 0.5 | 0.494 | 0.674 | 0.675 | 0.665 | 0.500 | +0.171 (0.080) | -0.009 (0.038) |

and the two cannot be separated here. Area under the curve on the batch ranged from 0.81 to 1.00 across the source divisions, so the model's ranking does not announce the failure either.

# S11 A batch the model chose

The library sample and the representation are those of Section S14. The 200 000 analysed molecules are divided once under seed 777 into 80 000 for training, 30 000 for calibration, 20 000 reserved from the planner and a screening pool of 70 000. The model is fitted on the training molecules with the standard hyperparameters and frozen; it is never refitted, so nothing in the campaign moves except which candidates have been committed. Its area under the curve on the pool is 0.966. Thresholds are set once on the calibration set at $\alpha = \delta = 0.10$ with the batch correction, and audit intervals use $\gamma = 0.05$.

In each round the planner commits the $m$ highest-scored candidates remaining in the pool. A quarter of the committed batch is drawn at random as a slice, under seed $5000 + r$, and the remaining three quarters are the batch to be priced. The same batch is priced three ways, so that only the audit differs: by a random sample of the pool still uncommitted, of the same size as the slice; by the slice itself; and by applying the same top-$m$ rule to the reserved pool, whose selected candidates are then removed from it. The committed candidates leave the pool and the next round begins. Tie-breaking is under seed $31337 + r$. Two campaigns were run, at $m = 8000$ for four rounds and $m = 1500$ for eight, giving twelve rounds in all. An arrangement is reported only where both its audit and the batch hold at least one in-class candidate.

The library audit could be formed in three of the twelve rounds and covered 2, 2 and 4 of the six consequence lines. The batch slice could be formed in nine and covered all six in every one. The reserved selection could be formed in ten and covered all six in eight of them, four of six in the other two. The rounds in which the batch was rich enough for the comparison to carry weight are the first round of each campaign and the second of the smaller one, at batch in-class rates of 0.269, 0.055 and 0.020; there the library audit was unavailable or covered 2 of 6, the slice covered 6 of 6 throughout, and the reserved selection 6, 4 and 6.

Depletion is recorded per round. At $m = 1500$ the pool fell from 0.0071 to 0.0013 after one round, then to 0.0008, 0.0007 and 0.0005, and stayed at 0.0005 while no arrangement could be formed from the fifth round on. At $m = 8000$ one round took it to 0.0005 directly. The rounds after the first are therefore priced on an exhausted library, where

a batch holds a handful of in-class candidates and most consequence lines are bounded near zero; their coverage is reported for completeness and carries little weight, which is why the three rich rounds are named separately above.

## S12 A pilot of the screened source, divided non-randomly

The divisions of Section S10 establish that provenance costs coverage; this asks whether the remedy the paper recommends works, under a division that could fail. The formulations are reduced as elsewhere, the room-temperature window and the $8\,\mathrm{mS\,cm^{-1}}$ cut applied unchanged. For each source contributing at least 40 room-temperature formulations, the other sources are divided 50/25/25 into training, calibration and an off-source audit; the model is fitted on every temperature row of the training formulations, with the standard hyperparameters, and frozen. The thresholds therefore come from formulations of other papers in every arm, which is the position a campaign is in, and only the origin of the audit changes. Three repeats were run per source under seeds $4242 + r$, with the division drawn under $909 + r$ and the tie-breaking jitter under $31337 + r$.

Five arrangements price the held-out source. *Off source* audits it with the held-back quarter of the other sources. *Random* divides the held-out source at random, which is exchangeable by construction and is reported as the ceiling. *By concentration* lets the low-concentration half audit the high-concentration half. *By system* partitions the source's salt–solvent systems so that none is shared. *As reported* lets the first half of the source's formulations in file order audit the second, which is not random but is chemically arbitrary, and so separates a division that is merely not random from one that is shifted in chemistry. An arrangement is run only where each side holds at least 20 formulations and at least one of each class, which admits eight sources for the off-source and random arms and six, three and five for the three genuine divisions.

Coverage of the six consequence lines, with the source as the unit, was $0.55 \pm 0.07$ off source, $0.92 \pm 0.17$ by concentration, $0.87 \pm 0.21$ by system, $0.86 \pm 0.12$ as reported and $0.99 \pm 0.02$ at random. On the sources admitting both, the paired gain of a pilot over an off-source audit is $+0.37 \pm 0.21$, $+0.37 \pm 0.26$ and $+0.34 \pm 0.17$. Median audit-to-batch in-class rate ratios were 0.56 off source, 1.04 by concentration, 1.09 by system, 1.81 as reported and 0.95 at random.

## S13 Label-noise sensitivity of the solubility screen

The solubility label is an indicator that the measured $\log S$ is at least $-4$, and AqSolDB pools sources whose measurements of the same compound typically differ by 0.5 to 0.7 log units. Taking 0.6 as the scale, the expected flip rate at the cut is 8 to 9% among the lipophilic compounds screened, 15% of which lie within half a log unit of it. The sweep is read at a resolution of the same order, so the analysis of Section 3.4 was repeated with the model, its hyperparameters, its scores and the tie-breaking jitter held exactly as published, changing only the labels.

Seven labellings were compared over all ten splits, 110 evaluations in all. *Base* is the published labelling, which reproduces the published split-0 figures exactly (144 distinguished settings, 104 on the Pareto front, 26 certifiable, enrichment $2.40\times$ at $p = 1.8 \times 10^{-15}$) and so validates the harness. *Cut* moves the threshold to $-3.5$ and to $-4.5$. *Random* flips 8% of the compounds within half a log unit of the cut, uniformly, under five seeds. *Disagreement* flips the same number in decreasing order of the recorded inter-source standard deviation, which is non-zero for 21.4% of compounds and gives a median flipped disagreement of 0.82 log units; this arm is asymmetric by construction, 54 of 61 flips running from insoluble to soluble, and raises the screened prevalence from 0.232 to 0.294. *Structured* flips those the score ranks most extremely against their class, and *adverse* its mirror image, those it ranks least extremely.

The distinguished-setting count stays between 132 and 144 in every arm but the upward cut shift, where its median falls to 120. The enrichment of the confident calls, with the setting held at the published request, lies between 1.51 and 3.61 and clears the Bonferroni level $0.05/144$ in all 100 evaluations where it is defined. The Pareto front holds its size, a median 102 to 114 settings, while retaining a median 74 to 85% of the base front's members and 44% in the worst split. The scenario degrades furthest: the median number of certifiable settings falls from 26 to 18 (random), 2 (adverse) and 0 (disagreement, with six of ten splits admitting none), and the cheapest certifiable budget rises from a median 134 per 1000 to 294, 380 and 378 respectively.

The structured arm is the one worth dwelling on, because it runs opposite to the mechanism of Section S15: flipping the compounds the score ranks most extremely raises the certifiable count to 66 and the fold to 3.18 and lowers the budget to 36. Noise correlated with the score makes the recorded labels agree with the model, which flatters every statistic the screen reports. The damaging direction is the adverse one. Score-structured label noise is therefore not ruinous in itself; its sign decides, and a proxy that agrees with the model conceals rather than reveals.

**As the main text had it.** The solubility labels are a cut through a continuous property measured across sources, so they are noisy where it matters, at the cut. Taking the spread between sources as its scale, about 8% of the screened compounds near the cut would be labelled the other way by another laboratory, which is the size of the requests being

swept; a sweep read at that resolution has to be shown to survive it. We therefore re-ran the whole analysis on the same model and the same scores, changing only the labels: moving the cut half a log unit each way, flipping 8% of the compounds within half a log unit of it at random, flipping the same number in order of their recorded disagreement between sources, and flipping those the score ranks most extremely against their class.

Two of the three claims survive and one does not. The resolution of the sweep survives: the number of distinguished settings stays between 132 and 144 under every perturbation but the upward shift of the cut, which lowers it to a median 120. The enrichment survives in sign and in significance, at a fold between 1.51 and 3.61, significant after the Bonferroni correction in every one of the hundred perturbed evaluations — so an enrichment of about twice the base rate is safe, though the two digits of 2.4 are not. What does not survive is the scenario, which is the part a team would plan on. The number of certifiable settings falls from a median 26 to 18 under random flips and to 2 and 0 under the two adversarial variants, with six of ten splits admitting no certifiable setting at all once the most discordant labels are flipped, and the cheapest certifiable budget rises by a factor of 2.2 to 2.8. The Pareto front keeps its size but not its membership, retaining a median 74 to 85% of the settings it held under the recorded labels and as little as 44% in the worst split. A quarter of the front is therefore an artefact of the labels, and a scenario chosen at this audit size and this label quality should be read as a demonstration of the machinery rather than as a plan.

One detail of this is worth stating because it cuts against the intuition the proxy-label section below establishes. Flipping the compounds the score ranks most extremely *flatters* the screen — more settings become certifiable, the enrichment rises, the budget falls — because noise correlated with the score makes the labels agree with the model. The damaging variant is its mirror image, flipping the compounds the score ranks least extremely against their class, and that is the one reported above. Score-structured label noise is therefore not uniformly ruinous: which way it is structured decides whether it hurts, and a proxy that agrees with the model is the one that hides its own failure.

# S14 The DNA-encoded library screen

**Data.** The library is BELKA,[26] released by Leash Biosciences through the Polaris record `leash-bio/belka-v1` under CC BY-NC-SA 4.0. Each molecule is a trisynthon: one building block at each of three positions on a shared core, with the DNA tag replaced by a placeholder atom in the released SMILES. The release carries a binary binding call per molecule for three targets; we use soluble epoxide hydrolase. A sample of 1 000 000 distinct molecules was drawn from the 98 807 372 molecules that carry labels for all three targets, by ranking molecules on the MD5 digest of the UTF-8 bytes of the standardized SMILES, the byte `0x1F` and the decimal seed 20261028, and keeping the lowest million, so the sample is reproducible from the release and is not enriched for binders; the release's own `split_group` partition was not used to choose molecules. In the million-molecule sample 7416 molecules bind the target, a rate of 0.74%, and the first, second and third positions hold 306, 754 and 935 distinct blocks. The analysis runs on a further random subsample of 200 000 of those molecules, drawn once under a fixed seed and used for every design and repeat; it holds 1442 binders, a rate of 0.72%, and 294, 716 and 899 blocks at the three positions. In-class molecules are strongly concentrated by block: the five richest first-position blocks carry 60.5% of the binders in the million-molecule draw and the richest binds at a rate of 0.709; in the 200 000 analysed the same five carry 61.1% and the richest binds at 0.685, against that sample's rate of 0.0072. The raw sequencing counts, replicates and controls live in a separate archive that was not downloaded, and the release's proprietary continuous scores were not used: the label is the released binary call. The release also carries its own held-out partition, tagging molecules `train`, `share` or `non_share`; because the seeded million-molecule draw is almost entirely `train`-tagged, that partition plays no part here and the designs below are our own. A consequence worth stating is that the released binary call is itself derived from selection counts, which is the kind of proxy label Section S15 shows can be ruinous when its errors are structured with respect to a model's score; enrichment calls are known to favour particular physicochemical classes, which a fingerprint model will learn, so the coverage results of this library are statements about the recorded call and not about confirmed binding.

**Why counts are not per 1000.** At an in-class rate below one percent a batch of 1000 contains about seven in-class molecules, and a setting that confirms a handful of hits then has a lower bound of zero whatever the audit size, so the outcome table carries no information. Intervals for this library are therefore quoted for the realized batch of each repeat, which runs to tens of thousands of candidates; the headline position-1 repeat has a batch of 42 145. The scripts additionally scale each rate to a nominal batch of $10^5$ so that repeats can be compared, and those scaled figures are not the intervals quoted in the main text. The thresholds were set with the batch correction at a planned $m = 1000$, inherited from the shared analysis code; the correction is conservative in $m$ and the deployed in-class order statistic was the 106th of 108, so the rule is more conservative than the request rather than less, but the mismatch between the planning scale and the reporting scale should be removed.

**Splits over blocks, not molecules.** A molecule's nearest neighbour in a combinatorial library differs by one building block, so assigning molecules at random leaves a held-out molecule one substitution from a training molecule, and measures interpolation rather than transfer. Designs were therefore defined over blocks. In *block1*, *block2* and *block3*,

the distinct blocks at the named position are partitioned between training, calibration, audit and batch, so a batch molecule carries a block at that position that no training molecule carries. Holding out all three positions at once is stricter but discards the 91% of molecules whose blocks disagree about their role and leaves too few in-class molecules to calibrate, so it is not reported. The *random* design assigns molecules at random and is reported alongside to show how much the easier split flatters the result. The position-1 and position-2 designs were also run with a uniform random score through the identical pipeline, as the null the enrichment test must reject and as a check that coverage and interval width survive a worthless score.

**The declared round.** A round was declared as a batch of which 30% is built on first-position blocks that the model has not seen, and three preparations were compared, each with its own calibration and audit sets and all three sharing one fitted model per repeat. The naive and mirror arms are priced on one and the same batch; the pilot's batch draws its novel share from all of the new blocks rather than from the half the other two use, so it is a different draw in every repeat and between 1% and 9% smaller in four of the ten. The pilot arm also splits a fixed pool at the molecule level, which makes its calibration, audit and batch exchangeable by construction and its result a positive control. *Naive* draws calibration and audit only from blocks the model trained on. *Mirror* gives them the same 30% novel fraction as the batch but draws that fraction from blocks disjoint from the batch's novel blocks, which is the position of a team that has measured some new chemistry and is about to make different new chemistry. *Pilot* gives them the same novel fraction drawn from the batch's own novel blocks, split at the molecule level, which is a pilot plate of the chemistry the round will contain. Ten repeats were run for each, and the comparison in the main text is of the six consequence lines of the default setting.

**The $\pm$ convention.** Coverage figures of the form $0.97 \pm 0.04$ are the mean over repeats plus or minus twice the standard error of that mean, with the repeat as the unit, not a standard deviation over repeats. Where a design contributes more than one run per repeat, the runs of a repeat are not independent and the repeat is still the unit.

**Model, thresholds and repeats.** Features, model and hyperparameters are those used throughout (Section S9); the model is refitted once per repeat and frozen before any threshold is set. Scores were tie-broken at random before thresholds were taken, which matters here because many molecules receive identical scores. Thresholds used $\alpha = \delta = 0.10$ with the batch correction, and audit intervals $\gamma = 0.05$. The block holdouts ran ten repeats each and the random design nine. Runs are appended to disk as they complete, so a long campaign can be run in chunks.

**The pre-flight check.** The label-free test of Section 3.10 trains the same model class on fingerprints alone to separate one set of molecules from another, with the molecules of each set capped at 12 000 and drawn at random, half for fitting and half for the reported area under the curve. Alongside it we report, for each of the three positions, the share of batch molecules whose block at that position also occurs in the comparison set. Three repeats were run per design.

The null is obtained by permutation. The audit and batch indices are pooled, split at random into two halves of the same sizes, and the same fit-and-score procedure is applied, twenty times per repeat at the same cap as the statistic itself; the statistic under that null is what a division with nothing to find returns at this sample size. Run on the three preparations of the declared round, over two repeats each, the audit-against-batch area under the curve was 0.658 (0.656–0.659) for the naive preparation, 0.663 (0.658–0.667) for the mirror and 0.502 (0.499–0.505) for the pilot. The permutation null averaged 0.500 over 120 replicates and spanned 0.491 to 0.511. An earlier run drew the null at a quarter of the statistic's own cap, which inflated its spread to 0.470–0.524 and so made a reading near one half easier to obtain; the figures here are from the corrected run, and the verdicts are unchanged. The two preparations whose intervals fail are therefore separated from the null by a wide margin and the one that holds is not, at a block share of 0.70 in the failing arms rather than the 0.00 of the block holdouts, so this is the design on which the test is informative rather than determined by construction.

**Diagnostics that do not order the failures.** Over the thirty block-holdout repeats the distance of the audit's in-class rate from the library's predicts how many consequence lines are covered only moderately (Spearman $-0.38$, $p = 0.04$), and within the ten first-position repeats alone it predicts nothing ($-0.05$). The distance between the audit's rate and the batch's predicts it well ($-0.68$ pooled, $-0.90$ at the first position), and needs the batch's labels. The enrichment of the confident calls over the audit's own rate runs 8.7–10.6 across the nine random-assignment repeats and 1.4–17.5 across the ten first-position repeats; it is significant in every repeat of every design, including the repeat at an enrichment of 1.4, because the batch holds tens of thousands of candidates.

**The widest overshoot.** In the first-position repeat whose promise overshot by the widest margin — not the worst by coverage, which is the repeat covering none of its six lines — a batch of 42 145 candidates was promised between 578 and 723 confident hits at a model AUROC of 0.89 and returned 119. That batch held 150 in-class molecules in

all, so the interval was already refuted by the two in-class rates, 1.62% in the audit against 0.36% in the batch, before the model was consulted. The same audit prices a nominal batch of 1000 at 8 to 24 confident hits, a separate interval computed at that size rather than this one rescaled.

**Block concentration.** In-class molecules are strongly concentrated by block: five first-position blocks carry 60.5% of the binders in the million-molecule draw, the richest binding at 0.709; in the 200 000 analysed the same five carry 61.1% and the richest binds at 0.685, against that sample's rate of 0.0072.

**Why the pilot arm's interval is narrower.** The pilot's audit holds 24 349 compounds against 17 045 and 17 315 for the other two arms, and 174 in-class against 120, because it draws its novel share from all of the new blocks rather than from half of them. Its interval on confident hits spans 70 candidates against 77 and 76, and the width formula of Section S3 accounts for essentially all of that difference: a predicted ratio of 0.911 against an observed 0.900. At a matched audit size the three widths would be indistinguishable, so the valid arrangement cost about 7300 extra audit measurements rather than buying a sharper interval. The pilot's batch is also a different draw in every repeat and, where the pool of new blocks runs short, between 1% and 9% smaller in four of the ten.

# S15 Calibrating against a proxy label

**The locality, in the terms a campaign uses.** One case is free, and for the reason that makes the construction rank-based. No part of the rule reads the numerical value of a score: a threshold is an order statistic of the calibration scores of one class, a prediction set follows from comparisons against it, and every count in the outcome table is a function of ranks. So the table is unchanged by any strictly increasing transformation of the score, which may therefore be a potency, a logit or an arbitrary monotone rescaling rather than a probability; we verify this on the library screen, where three such scales give tables agreeing cell for cell (ESI Section S15). The same argument applies to the label. If the proxy is an increasing function of the affinity, then being above a proxy cut-off and being above the corresponding affinity cut-off are the same event, so the two labels coincide and nothing in the analysis moves.

That the requirement is local can be made exact, and in terms a campaign already uses. A team picks a quantile above which it will make compounds, and the rule here is of that shape, with one cut per class: at a request of $1-\alpha$ the out-class cut sits near the $1-\alpha$ quantile of the out-class scores and the in-class cut near the $\alpha$ quantile of the in-class scores, measured at 0.93 and 0.07 in these runs. Holding the relabelled share at a tenth of each class, so that the confusion matrix is the same in every condition, and moving only where in the within-class ranking those compounds are taken from, the damage follows the cuts and separates by class. Mistakes placed at the out-class cut displaced the out-class distributions by 0.036 and the in-class ones by 0.007; placed at the in-class cut the figures exchange, 0.007 and 0.036; placed anywhere between the two they left both at 0.007 to 0.009, which is the sampling floor. Coverage shortfalls track this, 54 of 120 class-repeats at a class's own cut against 6 to 19 elsewhere, and the audit intervals again covered 96.4% throughout. A proxy must therefore order compounds correctly in a narrow window around the quantile the campaign has chosen, and is free to misorder them everywhere else: the requirement attaches to a number the team states, not one it has to discover.

**The shortfall as a gap between two conditional distributions.** Write $u$ for the frozen score, $y$ for the label a campaign cares about and $\tilde{y}$ for the proxy it calibrates against. The out-class threshold is the $k_0$-th order statistic of the scores of the calibration compounds with $\tilde{y}=0$, so it sits at the quantile $k_0/(n_0+1)$ of the score distribution conditioned on $\tilde{y}=0$, and that quantile is what the coverage request buys, up to the overshoot the batch correction introduces: $F(\tau_0 \mid \tilde{y}=0)=\ell$, which exceeds $1-\alpha$ and which the control measures at 0.924 against a request of 0.90. Carrying $\ell$ rather than replacing it by $1-\alpha$ is what makes Eq. 1 of the main text a three-term decomposition rather than a subtraction. The coverage the campaign actually obtains is measured on the compounds with $y=0$, and is $F(\tau_0 \mid y=0)$. Subtracting gives Eq. 1 of the main text. The in-class statement is the mirror image, with the survival function at the cut $1-\tau_1$ conditioned on $\tilde{y}=1$ and on $y=1$.

This is an accounting identity rather than a theorem, and that is the point of writing it down: it names the quantity a proxy has to control. The right-hand side involves the two conditional distributions only at $\tau_0$, so a proxy is free to disagree with the truth anywhere the score does not place a threshold, and the number of compounds it disagrees about enters nowhere. A proxy whose errors are independent of $u$ leaves both conditional distributions equal to the pooled one, up to the sampling error of the relabelled counts, and costs nothing. A proxy whose errors are concentrated at one end of the ranking shifts one distribution relative to the other and costs the difference.

**Data.** Selection counts and measured inhibition constants for carbonic anhydrase II, IX and XII were taken from the CA-DEL release[27] (Zenodo record 17656024, CC BY 4.0), whose affinity values derive from ChEMBL[28] and are distributed under CC BY-SA 3.0. No molecule in the release carries both a selection count and an affinity, so the two

tables cannot be joined; the proxy studied here is therefore constructed from the affinities rather than taken from the selection counts, which is what allows the structure of its errors to be controlled while the compounds and the score are held fixed. Molecules were standardized as elsewhere in this work and merged on InChIKey. Compounds whose reported affinities span more than one log unit were excluded as inconsistent (121, 0 and 16 compounds), leaving 3844, 3103 and 2160 compounds for the three isoforms. In-class means $\mathrm{p}K_i \geq c$ for $c \in \{7, 8\}$.

**The frozen score.** A regressor was trained on the selection counts of the library molecules, with the logarithm of one plus the count as its target, a tenth of the molecules held out, and the same representation and hyperparameters as elsewhere. Its predictions on the affinity-labelled compounds were mapped to $(0, 1)$ by a logistic transformation of the prediction standardized by the mean and standard deviation of the training molecules' predictions, so that no affinity enters the score. The score is frozen: it is the same in every condition of the experiment and does not depend on which proxy is under test. Its rank correlation with affinity is 0.22, 0.24 and 0.27 for the three isoforms, which is poor, and deliberately not repaired, because the experiment concerns the calibration label and not the model.

**Proxy families.** Each repeat draws a proxy label vector over all compounds and then splits them 30/30/40 into calibration, audit and batch. Thresholds are set from the calibration compounds using the proxy's labels at $\alpha = \delta = 0.10$ with the batch correction; the audit and the batch are then scored with the true labels. Three families are compared. *Increasing in affinity* thresholds a strictly increasing function of the affinity at a cut chosen to match the true prevalence, which returns the true labels themselves and is the control; it did so in every repeat. *Mistakes at random* relabels a fraction $f$ of the compounds of each class, chosen uniformly at random within the class. *Mistakes by score* relabels the same number of compounds of each class, choosing within the out-class those the score ranks highest and within the in-class those it ranks lowest, with ties broken at random. Because the two relabel identically many compounds of each class, their confusion matrices against the truth are identical, and so are their sensitivity, specificity, area under the curve and implied prevalence; the prevalence shift agreed to machine precision in all 18 matched pairs. Twenty repeats were run per isoform, cut-off and condition, with seeds fixed as a function of those indices.

**Results beyond Table 8.** Taking the gap of Eq. 1 as measured in each run and adding it to the control's coverage predicts the coverage achieved to within 0.0040 on average and 0.0106 at worst over all 36 conditions, the two classes counted separately. Audit-set intervals covered 4797 of 5040 consequence lines, 95.2% against a nominal 95%, with no trend across the conditions. A rank correlation between the proxy and the affinity does separate the two families slightly, 0.458 against 0.496 at $f = 0.20$ pooled, but in the direction opposite to the damage: the structured proxy correlates marginally better with affinity and covers twenty points worse.

**Where in the ranking the mistakes sit.** A second sweep holds the relabelled share fixed at a tenth of each class and moves only the position from which those compounds are drawn. Within each class the compounds are sorted by score and a contiguous window of the required size is taken, centred on a target position in the within-class ranking; positions 0.05, 0.10, 0.20, 0.35, 0.50, 0.65, 0.80, 0.90 and 0.95 were used, at both cut-offs and all three isoforms, 20 repeats each. Because the window size does not depend on its position, the number of compounds relabelled in each class is the same in every condition: the measured disagreement with the truth was 0.100 in all 54 conditions, so sensitivity, specificity, area under the curve and implied prevalence are constant and only the location of the mistakes varies.

Averaged over isoforms and cut-offs, the mean absolute value of the gap of Eq. 1 was 0.036 for the out-class and 0.007 for the in-class when the window sat at positions 0.90–0.95, 0.007 and 0.036 when it sat at 0.05–0.10, and 0.009 and 0.007 when it sat between 0.35 and 0.80. Coverage shortfalls were 54 of 120 class-repeats for the out-class at position 0.95 and 54 of 120 for the in-class at position 0.05, against 6 to 19 of 120 at the intermediate positions. The dissociation is therefore by class as well as by position: a class is perturbed by mistakes near its own cut and not by mistakes near the other's. The sign of the perturbation is not fixed, because whether relabelling displaces a cut upwards or downwards depends on which compounds leave and join the class; over-coverage wastes budget where under-coverage breaks the guarantee, and both indicate that the cut is no longer where the request put it. Audit-set intervals covered 6247 of 6480 consequence lines, 96.4%, with no trend across positions.

**Monotone rescaling of the score.** That the outcome table depends on the score only through its ordering was checked on the library screen of Section S9. Ties were broken once on the original scale, and the whole analysis was then run on the model's probability of being in-class, on its logit, and on a potency-like scale obtained from the logit by an affine map composed with an inverse hyperbolic sine. All three gave outcome tables agreeing in every cell, with identical published intervals and identical verdicts, and on each scale both cut points were confirmed to be order statistics of the calibration scores of their own class. The consequence for reporting is that a threshold may be quoted on whichever scale a reader can interpret, a potency rather than a probability of a binary label, at no cost in coverage, provided the map is strictly increasing; and the mapping may be fitted on any data at all, including the calibration labels, because the rule never reads the values it produces.

## S16 Software and scripts

All analyses reported in this paper were run with the scripts below, which implement the methods directly with NumPy and SciPy; they do not depend on a separate package. The repository holds further scripts from exploratory work that this paper does not report, covering identifiability in combinatorially generated libraries, selectivity endpoints and presentation material; they are listed in the repository's own index and are kept so that the record is complete rather than because any result here rests on them. The author's open-source package `ssbc` provides the threshold corrections, outcome tables and audit-set intervals for general use, under the MIT licence. [ssbc version, URL and archived DOI, on acceptance.] Library versions are listed with the scripts. Every script is seeded.

- Molecular pipeline: `featurize.py` (standardization, fingerprints, scaffolds), `conformal.py` (thresholds, cells, tie-breaking), `rowint.py` (calibration-only intervals), `run_tox21_v2.py` and `run_aqsol.py` (model fits; scores, labels and split indices stored), `analyze_tox21_v2.py` and `analyze_aqsol.py` (all molecular numbers, including the Poisson-binomial tests, pilot statistics, widths and monotonicity), `check_batch_law.py` (Table S11), `make_figures_mol.py` (main text Figs. 4 and 3);
- `calisol.py`, `calisol_summary.py`: electrolyte screen (main text Table 6, Fig. 5);
- `scenario_map.py`: scenario maps (main text Fig. 3, Fig. S3);
- `audit_check.py`: real-data check of audit intervals and of choosing a setting on the audit set (main text Table 3);
- `beyond_chance.py`: interval width against the trivial interval, and the Fisher test of enrichment of the advanced set (main text Sections 3.4 and 3.6);
- `plate_bounds.py`: audited consequences of the new-salt designs, intervals for one 96-well plate, and the joint measurement bound at other audit sizes (main text Sections 3.4 and 3.6);
- `intervals.py`: prediction limits used by all scripts (conventions of Section S3);
- `threshold_sim.py`: threshold correction (Table S1);
- `validate_intervals.py`, `finite_library.py`, `cover2.py`: interval validation (main text Fig. S5, Tables S6, S5 and S7);
- `calib_only_rowsim.py`: calibration-only intervals (Table S2, Fig. S6);
- `decision_sims.py`: cost and scaffold simulations (Table S3, main text Table S3, Fig. S4); `selection_sim.py`: selection among rules (its fixed-order variant is not reported, for the reason given in Section S7); `ties_sim.py`: tied scores (Table S4);
- `belka_screen.py`: the block-holdout screen (main text Section 3.8, Section S14); the seeded million-molecule sample is specified by the digest rule of the Data paragraph above rather than by a script, and is reproducible from the release without one; `stratified_blocks.py`: the attempt to equalize prevalence across block-disjoint roles;
- `next_round.py`: the declared round and its three preparations, naive, mirror and pilot (main text Section 3.8);
- `exch_check.py`: the label-free pre-flight check and the block shares (main text Table 7); `preflight_arms.py`: the same check on the three preparations of the declared round, with its permutation null;
- `cadel_fetch.py`, `surrogate_structure.py`, `where_mistakes.py`, `monotone_check.py`: the carbonic anhydrase proxy-label experiments and the monotone-rescaling check (main text Table 8, Section S15);
- `sizing.py`: guaranteed-yield sizing and the audit-size inversion (main text Fig. 2, Section S3);
- `adaptive_campaign.py`: the campaign in which the planner commits the top of the ranking (main text Section 3.9, Section S11);
- `pilot_nonrandom.py`: the pilot of the screened source under non-random divisions (main text Section 3.7, Section S12);
- `aqsol_noise.py`: the label-noise sensitivity of the solubility screen (main text Section 3.4, Section S13);
- `source_split2.py`: the division of the electrolyte formulations by provenance (main text Section 3.7, Section S10). An earlier `source_split.py` fitted the model on its own calibration set and is superseded; its figures are not the ones reported;

- `within_between.py`, `stereo_blocks.py`, `support_check.py`: the intracluster correlations and design effects, the stereoisomeric block pairs, and the similarity-cluster counts of the Limitations;
- `thatcher.py`: numerical check of the hypergeometric–Beta–Binomial identity;
- `make_figures.py`, `make_figures_rev.py`, `dkw_and_map.py`: remaining figures; `make_tables_rev.py`, `make_tables_v2.py`: all tables, written from stored results.

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